The Entropic Scree: An Information-Theoretic Diagnostic Framework for Intrinsic Rank and Informational Gravity in Tabular Systems

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Abstract

Determining intrinsic dimensionality is foundational for manifold learning and predictive modeling. However, standard methods relying on linear variance or spatial distance metrics require strict assumptions that rarely hold in complex, high-dimensional environments, often fragmenting dependencies and driving spurious dimensional inflation. This paper introduces the Entropic Scree, an information-theoretic diagnostic framework evaluating intrinsic dimensionality via the eigendecomposition of Normalized Mutual Information. Using Information-Theoretic Jaccard Similarity (Variation of Information), it produces a geometrically coherent metric space unifying continuous, ordinal, and discrete variables, compressing fragmented dimensions back towards their Intrinsic Generative Rank. Crucially, the method maintains rigid structural boundaries under root entanglement, where baselines like Kernel PCA systematically degrade. Evaluating logarithmic spectral decay separates primary generative drivers from the Extended Signal Tail and unstructured idiosyncratic informational variance. Additionally, the framework quantifies the total signal-to-idiosyncratic volume and computes Average Informational Gravity (AIG) and Factor-Specific Informational Gravity (FSIG) metrics to translate abstract matrix properties into actionable, variable-equivalent footprints. Furthermore, bipolar eigenvector loadings and the Dimensional Inflation Index can be used to identify decoupled sub-networks and assess a dataset's linear sufficiency, respectively. Ultimately, the Entropic Scree provides a non-parametric, distribution-free diagnostic whose comparative advantages compound at scale and with system complexity.

Keywords: Information Theory; Principal Component Analysis; Factor Analysis; Sklar's Theorem; Copula Theory; Non-linear Manifold Learning; Dimensionality Reduction

1 Introduction

The ability to accurately estimate the intrinsic dimensionality of a dataset – the true number of underlying structural drivers of the observed data – dictates the success or failure of downstream statistical modeling, clustering, and manifold extraction. If dimensionality is underestimated, true structural dependencies are permanently discarded. If dimensionality is overestimated, empirical estimators risk overfitting, capturing localized idiosyncratic informational variance that degrades the generalization of the true generative signal.

For decades, the universal diagnostic standard for evaluating representational rank has been Principal Component Analysis (PCA) and the visual or algorithmic inspection of its eigenvalue decay (the scree plot) (Cattell, 1966). PCA operates by computing a bivariate linear covariance or Pearson correlation matrix and projecting the data onto orthogonal axes that maximize variance. However, modern enterprise and scientific data environments often violate the parametric assumptions required for standard PCA to succeed. Real-world systems are characterized by heterogeneous data types (mixing continuous sensors with discrete categorical states), heavily non-linear and synergistic interactions, and dimensional sparsity where the physical breadth of recorded variables vastly outnumbers the available observations ($m \gg N$).

When applied to these environments, standard PCA does not merely suffer from reduced precision; it systematically produces misleading structural diagnostics that distort the true geometric complexity of the data. This paper outlines the specific structural mechanisms driving these linear failures and introduces the Entropic Scree – an information-theoretic methodology designed specifically for complex, mixed-data systems. It serves as a replacement for traditional variance-based metrics and other dimensionality estimators constrained by assumptions about the underlying generative model. While applicable to any mixed-type dataset, its comparative advantage over established extraction paradigms compounds in high-dimensional environments, where the sheer volume of non-linear combinations structurally fractures standard spectral matrices.

When a dataset is strictly continuous and perfectly generated by a linear multivariate Gaussian model, standard PCA is mathematically optimal. Under these pristine, parametric conditions, mutual information becomes a direct function of linear correlation, and the Entropic Scree method introduced in this paper converges with classical PCA rank estimations. The power of the Entropic Scree does not lie in overriding PCA within its native domain, but in its structural independence. It relies on no assumptions regarding the linearity, continuity, or parametric form of the underlying generative model, effectively subsuming classical factor theory while remaining robust in the flawed environments where classical models collapse.

1.1 Distinction from Topological Intrinsic Dimensionality

The structural rank extraction presented in this methodology is distinct from the broader field of Topological Intrinsic Dimensionality (ID) estimation. The machine learning literature offers a robust suite of non-linear ID estimators, most notably the Maximum Likelihood Estimator (MLE) (Levina and Bickel, 2004) and nearest-neighbor frameworks such as TWO-NN (Facco et al., 2017). While highly effective for continuous spatial manifolds (e.g., image or coordinate data), these topological estimators evaluate local neighborhood density via Euclidean or geodesic distance metrics between samples. Consequently, they are structurally incompatible with the ill-conditioned data regimes addressed by the Entropic Scree. As established by the phenomena of distance concentration, in highly asymmetric, feature-rich environments ($m > N$), the ratio between the nearest and farthest neighbor distances converges to one (Beyer et al., 1999; Aggarwal et al., 2001). This spatial concentration renders the local nearest-neighbor calculations required by TWO-NN structurally degenerate. Furthermore, relying on standard topological estimators in these sparse regimes is known to induce underestimation of the true structural rank due to the collapse of their uniform linear neighborhood assumptions (Ceruti et al., 2014; Gomtsyan et al., 2019). Additionally, computing continuous geometric distances across mixed-data margins (e.g., continuous waves versus discrete binary states) produces meaningless topological embeddings.

Similarly, non-linear spectral mappers like Kernel PCA (KPCA) project data into infinite-dimensional Hilbert spaces. Because the resulting eigenspectra lack the finite-sample boundaries defined by Random Matrix Theory, they offer no algorithmic mechanism (such as a BBP phase transition) to discretely separate generative rank from the spectral tail.

Therefore, classical PCA and its factor-analytic derivatives remain the comparative baselines for this methodology not as strawmen, but because they represent the standard paradigm for extracting discrete, global generative drivers in complex, high-dimensional, mixed tabular environments where local topological ID estimators systematically fail.

2 Structural Constraints of Linear Estimators

Deploying standard PCA in mixed-data, high-dimensional environments reveals structural limitations across four dimensions:

- **Mixed-Data Penalty:** Complex systems inherently mix continuous, binary, and ordinal data. When a Pearson correlation matrix evaluates a continuous wave against a discrete step-function (point-biserial correlation), the maximum possible correlation is artificially deflated. Fundamentally, the severity of this deflation is inconsistent – it is dictated by the arbitrary shapes of the individual marginal distributions (such as varying class imbalances) rather than the true deterministic relationship (Cohen, 1983; MacCallum et al., 2002). Consequently, the linear metric does not merely project a structurally weaker manifold; it projects a geometrically warped one, where relative distances are distorted by marginal shape mismatches rather than reflecting the true shared signal.

- **Non-Linear Blindness:** Standard PCA only measures linear relationships. If the true relationship between variables is non-linear (e.g., polynomial interactions or thresholded discrete states), the linear correlation may be near zero. The structural signal functionally escapes the matrix and is incorrectly subsumed into the idiosyncratic informational variance.

- **Algebraic Rank Constraint:** In environments characterized by high-dimensional predictor breadth (e.g., dense sensor arrays, genomics, or telemetry), the number of features frequently exceeds the number of historical observations ($m > N$). Under these conditions, standard PCA encounters a mathematical boundary. Because PCA embeds variables as linear vectors entirely within the geometric coordinate space defined by the available samples, the rank of the sample covariance matrix is strictly capped at $N - 1$. If a complex system is intrinsically driven by 500 topological axes but is observed through only 100 historical snapshots, PCA artificially truncates the manifold at 99 dimensions, collapsing the remaining 401 eigenvalues to exactly zero.

- **Spurious Orthogonalization and Dimensional Inflation:** This constraint is the direct consequence of non-linear blindness and fundamentally drives PCA's tendency to overestimate dimensional rank. According to Price's Theorem (Price, 1958), forcing a continuous signal through a discrete, non-linear step-function (e.g., thresholding a sensor into categorical states) inherently generates an infinite series of high-order polynomial interactions. If X is a continuous Gaussian generative driver, any non-linear discrete proxy $Y = f(X)$ can be formally decomposed via the mathematics underlying Price's Theorem into a series of orthogonal Hermite polynomials (H_k):

$$Y = f(X) = \sum_{k=1}^{\infty} c_k H_k(X) = c_1 X + c_2(X^2 - 1) + c_3(X^3 - 3X) + \dots$$

Because standard PCA only evaluates linear covariance, and because the polynomial expansions of a Gaussian variable are linearly uncorrelated ($\mathrm{Cov}(H_i(X), H_j(X)) = 0$ for $i \neq j$), the correlation matrix cannot recognize that these terms originate from the exact same root X. Instead, the matrix acts as a rigid linear filter. To span the inflated variance of these statistically uncorrelated terms, the linear eigensolver is structurally forced to interpret each polynomial degree as a completely independent, spurious orthogonal dimension. As the feature space expands, the observable non-linear interaction space naturally fills out, and standard PCA structurally maps the expanded discrete headcount of this active expansion space. It does not move closer to the underlying generative structure; it merely produces a higher-resolution map of a fragmented linear projection. Because this fragmentation multiplies combinatorially as predictor breadth (m) grows, the failure of classical matrices is not merely a loss of precision, but a fundamental structural collapse at scale.

3 Entropic Scree Methodology

The Entropic Scree resolves these four structural limitations not by altering the mechanics of eigendecomposition, but by transforming the analytical space in which it operates. To break the constraints of marginal shape mismatch and linear blindness, the framework replaces linear variance with Mutual Information.

This substitution is firmly anchored in copula theory. In classical linear statistics, measures of dependency (like covariance) are structurally confounded by the individual shapes of the variables being compared. By Sklar's Theorem (Sklar, 1959), any joint cumulative distribution function can be decomposed into its univariate marginals and a copula – a distinct function that binds them together. The copula isolates the pure, underlying dependence structure by neutralizing the marginal distributions via the Probability Integral Transform. Because Mutual Information evaluates Shannon entropy (probability mass) rather than physical geometric distance, it is strictly invariant to these monotonic marginal transformations. Consequently, computing the mutual information of raw variables is formally equivalent to computing the exact information content of their underlying copula.

As established by Ma and Sun (2011), the mutual information of two variables equates directly to the negative differential entropy of their bivariate copula density ($c_{i,j}$):

$$I(X_i; X_j) = -h(c_{i,j})$$

While Sklar's theorem guarantees a unique copula for continuous variables, applying this framework to mixed-data systems introduces a theoretical boundary: for discrete variables, the copula is uniquely defined only on the Cartesian product of the observable marginal ranges, forming a *sub-copula*. However, this theoretical non-uniqueness in the continuous extension does not compromise the exactness of the shared informational volume. Because empirical mutual information evaluates probability mass strictly at these observable discrete coordinates, it maps the sub-copula without requiring artificial interpolation into the undefined continuous space.

By formalizing this link between mutual information and copula theory, the methodology projects continuous, ordinal, and categorical variables onto a unified, dimension-free probability scale. It evaluates the pure geometry of their dependence, rendering the resulting matrix structurally invariant to the marginal distortions that fracture classical correlation.

However, we acknowledge a boundary of constructing a 2D spectral space: because this matrix evaluates only bivariate marginal dependencies, it is structurally blind to pure higher-order multivariate synergies (such as strict XOR combinations) where the pairwise mutual information evaluates to zero despite the presence of joint deterministic relationships. This restriction to a pairwise framework is both a computational and algebraic necessity. Expanding mutual information estimators to higher-order combinations across thousands of variables ($O(m^k)$) is computationally prohibitive, and the mechanics of spectral eigendecomposition fundamentally require a two-dimensional ($m \times m$) similarity matrix to extract valid spatial coordinates. Therefore, this bivariate blindspot is not a unique failure of information theory, but a constraint shared by all *spectral dimensionality estimators*, including standard, rank, and Kernel PCA. Consequently, the proposed method evaluates the complete network of overlapping bivariate links, but leaves the extraction of pure multi-way combinatorial spaces to the downstream non-linear manifold algorithms (related discussion provided in Section 6).

3.1 Normalized Information Matrix (Variation of Information)

While raw Mutual Information $I(X_i; X_j)$ provides a pure measure of shared dependency, utilizing it directly to construct a similarity matrix for spectral analysis creates a structural imbalance in mixed-data systems. Because self-information equates to marginal entropy ($I(X_i; X_i) = H(X_i)$), the diagonal of an unnormalized matrix varies substantially based on variable cardinality. To prevent biased spatial embedding, the raw similarity matrix must first be standardized to a uniform unit diagonal ($\mathcal{M}_{i,i} = 1$). Without this initial normalization, continuous variables with large marginal entropies act as spectral gravity wells, suppressing low-cardinality discrete variables solely due to their measurement scale.

However, the specific mathematical mechanism used to achieve this unit diagonal dictates the geometric integrity of the entire space. Naive normalizations – such as dividing by the arithmetic mean ($\frac{H(X_i)+H(X_j)}{2}$) or the minimum entropy – offer a symmetric compromise, but both fail to satisfy the triangle inequality, allowing spatial contradictions to persist in the eigenspace. As an eigensolver attempts a Euclidean embedding, it compensates for these impossible local shapes by warping the dominant eigenvectors.

To guarantee global geometric coherence, the methodology employs Information-Theoretic Jaccard Similarity (Joint Entropy Normalization) to construct a Normalized Mutual Information (NMI) matrix:

$$\mathcal{M}_{i,j} = \frac{I(X_i; X_j)}{H(X_i, X_j)} = \frac{I(X_i; X_j)}{H(X_i) + H(X_j) - I(X_i; X_j)}$$

This formulation evaluates the shared informational variance (hereafter referred to as the structural *signal*) against the exact Joint Entropy of the pair. Because Joint Entropy penalizes variables for unshared idiosyncratic informational variance, this normalization imposes a conservative penalty on marginal divergence. Translating this specific normalization into a distance metric ($1 - \mathcal{M}_{i,j}$) yields the Normalized Variation of Information (NVI) (Rajski, 1961). NVI is a true metric space that obeys the triangle inequality, ensuring the extracted eigenvectors ($\mathbf{V}$) represent pure, orthogonal axes of the system's structural topology.

This entropic penalty must be distinguished from the linear mixed-data penalty (point-biserial deflation) discussed in Section 2. Standard PCA artificially deflates correlations due to a geometric shape mismatch – it attempts to fit a continuous straight line through a discrete step-function. In contrast, Jaccard normalization penalizes mixed data based solely on unshared informational volume. When a high-resolution continuous sensor interacts with a coarse discrete variable, the continuous variable contains excess structural entropy that the discrete proxy simply lacks the resolution to map. The entropic penalty is not an arbitrary algorithmic failure; it is an objective accounting of unshared uncertainty and lost resolution.

3.2 Double-Centering and Information Eigendecomposition

While the NMI matrix ($\mathcal{M}$) forms a valid metric space, performing eigendecomposition directly on raw similarities constitutes a geometric category error. Although eigendecomposition is algebraically executable on any real symmetric matrix via the Spectral Theorem, Warren Torgerson established that eigenvectors derived directly from a matrix of raw distances or similarities cannot be geometrically interpreted as genuine Euclidean spatial coordinates (Torgerson, 1952).

To extract a valid coordinate space, the underlying linear algebra requires constructing an inner-product (Gram) matrix defined around the dataset's center of informational mass (hereafter referred to as the centroid). Furthermore, empirical mutual information estimators exhibit a strictly positive finite-sample bias. By the Perron-Frobenius theorem for positive matrices (Horn and Johnson, 2012), if left uncorrected, this strictly positive estimation bias generates a disproportionate, artificial primary eigenvalue that artificially inflates the leading axis and obscures true structural gravity.

To satisfy the geometric requirement for a Gram matrix and neutralize this estimation bias, the proposed methodology executes a double-centering transformation:

$$\mathcal{M}_c = \mathbf{H}\mathcal{M}\mathbf{H}$$

where $\mathbf{H} = \mathbf{I} - \frac{1}{m}\mathbf{1}\mathbf{1}^T$ is the standard centering matrix. This single operation serves a dual mathematical purpose:

- **Geometric Perspective (Sub-Manifold Embedding):** It executes an advantageous variation of classical Multidimensional Scaling (cMDS). While standard Torgerson-Gower scaling constructs a Gram matrix by double-centering squared distances ($-\frac{1}{2}\mathbf{H}\mathbf{D}^{(2)}\mathbf{H}$), evaluating $\mathbf{H}\mathcal{M}\mathbf{H}$ algebraically simplifies to $-\mathbf{H}\mathbf{D}\mathbf{H}$ (where $\mathbf{D}_{i,j} = 1 - \mathcal{M}_{i,j}$). Consequently, the framework defines an inner-product space where the embedded squared Euclidean distance equals $2(1 - \mathcal{M}_{i,j})$. This signifies that the extracted coordinates embed the square root of twice the Normalized Variation of Information ($\sqrt{2 \cdot NVI}$). Because scaling and taking the square root of a valid metric preserves metric space properties, this geometry remains globally coherent. Furthermore, this specific square-root embedding is an established mechanism for elevating the Positive Semi-Definiteness of non-Euclidean distance matrices, minimizing synergistic spectral artifacts and ensuring a highly stable coordinate space (Bavaud, 2010; Schoenberg, 1938a).

In the context of the broader machine learning literature, this operation explicitly links the Entropic Scree to Kernel Principal Component Analysis (KPCA) (Schölkopf et al., 1998). By evaluating $\mathcal{M}_c = \mathbf{H}\mathcal{M}\mathbf{H}$, the framework effectively treats the Jaccard-normalized similarities as an empirical Gram matrix generated by a distance-based kernel. However, unlike standard Gaussian (RBF) kernels that implicitly project into an infinite-dimensional Reproducing Kernel Hilbert Space (RKHS) devoid of finite-sample boundaries, this information-theoretic kernel maps dependencies within a tightly bounded probabilistic volume. Because the underlying Normalized Variation of Information (NVI) accommodates synergistic dependencies, the centered matrix acts as an indefinite kernel space (Haasdonk, 2005; Ong et al., 2004). This theoretical alignment formalizes why the subsequent PSD projection (zero-clipping) is required: it acts as a mathematically optimal projection from an indefinite kernel space onto a valid Euclidean sub-manifold.

- **Statistical Perspective (RMT Bias Correction):** By forcing the rows and columns to sum to zero, double-centering algebraically mitigates the global mean of the positive finite-sample estimation bias, double-centering the macroscopic bulk of the idiosyncratic informational variance at zero. However, the finite-sample bias of empirical mutual information (e.g., the Miller-Madow correction (Miller, 1955)) scales proportionally with the number of discrete states of the interacting variables (specifically, $(|X| - 1)(|Y| - 1)$). Consequently, in mixed-data environments, the estimation error varies drastically across the matrix: the positive bias between two highly binned continuous sensors is significantly larger than the bias between two low-cardinality binary flags. Because this estimation error is heteroskedastic, double-centering neutralizes the uniform rank-1 global shift but cannot perfectly eliminate the pair-specific residual bias. This residual heteroskedasticity directly drives the heavy-tailed shape of the remaining idiosyncratic informational variance.

Post-double-centering, the matrix $\mathcal{M}_c$ no longer represents raw pairwise similarities; it maps topological information variance. The matrix diagonal is no longer locked at 1.0; instead, the strictly positive diagonal entries represent the squared geometric distance of each variable from the dataset's centroid, dictating the absolute structural weight (informational gravity) of the variable.

To counterbalance these positive diagonals, the zero-sum row constraint dictates that the net aggregate of the off-diagonals must be strictly negative. In this inner-product space, an individual off-diagonal entry defines the continuous geometric angle between two variables relative to the centroid. Because the double-centering transformation shifts the coordinate zero-point from absolute statistical independence to the pair's expected marginal baseline (a function of their individual connectivity to the broader dataset), these values must be interpreted relative to this dynamic threshold. A positive entry (an acute angle) indicates a shared dependency greater than their expected baseline, binding the variables into a localized structural neighborhood. An entry of zero indicates a shared dependency equal to their expected baseline. Lastly, a negative entry (an obtuse angle) indicates a shared dependency less than their expected baseline, mapping a structural estrangement that physically pushes the variables to opposite sides of the manifold. Crucially, negative entries should not be interpreted as inverse correlations, nor can positive entries be read as isolated factors. Instead, the matrix functions entirely as a global geometric blueprint from which the eigensolver extracts the true orthogonal axes.

Performing an eigenvalue decomposition on this centered inner-product matrix yields $\mathcal{M}_c = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$. Because the proposed methodology evaluates variables as non-linear probability distributions rather than linear vectors bounded by a sample-coordinate system, it successfully bypasses the algebraic rank constraint. The resulting $m \times m$ pairwise matrix aims to map the true topological overlap of the joint dependency structure, unconstrained by the algebraic rank ceiling of N. However, while it escapes the sample-size ceiling, by imposing the constraint that every row and column must sum to exactly zero, the double-centering transformation algebraically consumes one degree of freedom. This caps the active spectral rank at $m-1$ and guarantees at least one exact geometric zero, acting as the structural boundary between the positive Euclidean mass and the negative non-Euclidean artifacts.

By taking the eigenvalues ($\lambda_1 \geq \cdots \geq \lambda_m$) and zero-clipping any negative artifacts produced by non-Euclidean synergistic curvature ($\lambda_i^+ = \max(0, \lambda_i)$), the methodology executes the optimal projection onto the Positive Semi-Definite (PSD) cone (Higham, 1988). This projection does not discard any of the system's underlying structure, nor does it corrupt the primary axes. Spectral perturbation theory – specifically Weyl's inequality and the Davis-Kahan theorem (Davis and Kahan, 1970) – guarantees that the dominant positive eigenvectors containing the core structural signal remain stable under this optimal projection.

Mutual Information comprehensively captures all dependencies as positive probability mass. Furthermore, because Jaccard normalization enforces the triangle inequality (Section 3), these negative artifacts do not arise from impossible geometric contradictions. As established by Schoenberg's Theorem (Schoenberg, 1938b), they are simply the geometric obstruction to achieving a perfectly flat Euclidean embedding of a valid, curved geometry – a topological reality routinely managed in classical Multidimensional Scaling (cMDS) by isolating the positive spectrum (Torgerson, 1952). Finally, because the Entropic Scree delegates the physical construction of the manifold to downstream non-linear algorithms operating on the raw data (Section 6), the physical geometries of these synergies remain protected in the final feature space.

Because non-Euclidean synergistic curvature produces negative spectral artifacts, the sum of the clipped positive eigenvalues defines the system's total Constructive Spectral Mass ($m_+ = \sum_{i=1}^m \lambda_i^+$).

Furthermore, because double-centering shrinks the matrix trace to correct for estimation bias, the geometric baseline of the system is given by $Tr(\mathcal{M}_c)$. The difference $m_- = m_+ - Tr(\mathcal{M}_c)$ quantifies the absolute volume of the Synergistic Curvature Deficit ($|\sum \lambda^-|$) absorbed during PSD projection. To explicitly monitor topological collapse, we introduce the Synergistic Curvature Deficit Ratio (SCDR):

$$\text{SCDR} = \frac{m_-}{m_+} = \frac{\sum |\lambda^-|}{\sum \lambda^+}$$

This metric dynamically quantifies the percentage of topological variance discarded to achieve Euclidean flatness. If the SCDR exceeds acceptable engineering tolerances, it indicates the sub-copula possesses extensive non-Euclidean synergies that a flat PSD projection fundamentally cannot support, warning the practitioner of structural truncation.

3.3 Spectral Entropy and Total Probabilistic Volume

To construct a valid discrete probability distribution ($\sum p_i = 1.0$) for information-theoretic volume evaluation, spectral weights are normalized against this constructive mass:

$$p_i = \frac{\lambda_i^+}{m_+} = \frac{\max(0, \lambda_i)}{\sum_{j=1}^{m} \max(0, \lambda_j)}$$

Exponentiating the Shannon entropy of this normalized distribution ($R_{eff} = \exp(-\sum p_i \ln p_i)$) yields the dataset's Total Unique Probabilistic Volume. In classical information theory, this formulation defines the perplexity of a distribution – the effective number of equally likely states (Cover and Thomas, 2006). By applying perplexity directly to the spectral variance distribution, the framework translates abstract matrix mass into a highly interpretable count of *variable equivalents*.

We note that in classical linear statistics, covariance matrices are frequently standardized into correlation matrices (forcing the diagonal to exactly 1.0) to neutralize arbitrary physical units of measurement. However, because the uncentered NMI matrix was already constructed in absolute units of probability mass via Jaccard normalization, applying a classical correlation transformation post-centering structurally invalidates the metric space. Standardizing the matrix would artificially stretch purely noisy variables and shrink dominant structural anchors until all vectors possessed a length of 1.0, effectively giving idiosyncratic informational variance the exact same geometric voting power as the dominant true signals. Therefore, the matrix must remain in its pure double-centered inner-product form, preserving the exact informational gravity of every variable to allow the eigensolver to accurately map the dominant topological axes and preserve the absolute magnitudes required for R_{eff}.

3.4 Logarithmic Scale and Signal Boundary

Translating this continuous probabilistic volume into a discrete, physical dimensional count requires evaluating the topological geometry of the decay curve. Because the Entropic Scree evaluates information derived from Shannon Entropy ($H = -\sum p \ln p$), the geometry should be evaluated on its native logarithmic scale.

Evaluated on a linear scale, geometric curvature frequently estimates an elbow near the bottom of the primary spectral cliff, irreversibly discarding sparse, long-tail dependencies. Conversely, in high-dimensional tabular environments ($m \geq N$), finite-sample estimation error pools into the trailing eigenvalues. Because the NMI matrix bypasses the sample-size ceiling, this error forms a continuous probability distribution of unstructured estimation error. We explicitly note that the exact analytical forms of the Marchenko-Pastur (MP) law and the Baik-Ben Arous-Péché (BBP) phase transition (Marchenko and Pastur, 1967; Baik et al., 2005) are proven only for linear sample covariance (Wishart) matrices. We do not claim the double-centered, bounded, heteroskedastic NMI matrix obeys these exact analytical bounds. Rather, we invoke them purely as a macroscopic, phenomenological analogy: empirical non-linear similarity spaces frequently exhibit a structurally analogous localized idiosyncratic informational variance.

To safely extract the complete manifold, the diagnostic evaluates the decay on a logarithmic scale, defining $E_i = \ln(\lambda_i^+)$ for all $\lambda_i^+ > 0$. Exactly like Cattell's classical variance-based scree test (Cattell, 1966), the Entropic Scree is designed as a visual diagnostic framework. **The core methodological contribution of this paper resides in the construction of the valid non-linear spectral space and the translation of its mass into informational gravity, not in the automation of its parsing.** Human visual inspection of the log-linear spectral decay remains the gold standard for identifying the structural elbow separating generative signal from the idiosyncratic informational variance baseline.

When analyzing this log-linear decay, evaluating eigenvalue ratios $(\lambda_i/\lambda_{i+1})$ (Onatski, 2010; Ahn and Horenstein, 2013) rather than absolute variance drops helps resolve the scale imbalance between pervasive global drivers and subtle generative factors. Furthermore, topological and psychometric theory dictates explicitly bypassing the primary spectral gap $(i = 1)$, forcing a minimum structural extraction of $K \geq 2$. As observed by Cattell, the first dominant root in highly correlated systems typically absorbs a single "general factor" (a global magnitude gradient) (Cattell, 1966). Similarly, in classical scaling matrices, the first principal coordinate isolates global size rather than topological shape (Jolliffe, 2002). From a manifold perspective, a 1-dimensional extraction $(K = 1)$ is topologically degenerate, projecting the dataset onto a single line incapable of resolving complex interaction networks.

To support fully automated machine learning operations (MLOps) where human visual intervention is not feasible, the codebase provides an automated Dual-Diagnostic Ensemble heuristic that operationalizes these principles. Conceptually, it establishes a strict boundary by isolating the macroscopic idiosyncratic informational variance cliff. It then evaluates the remaining bounded signal space using the Maximum Secondary Eigenvalue Ratio (Log-Gap test) to identify the primary generative rank, alongside a localized geometric breakout test (Triple-Tap) to map the termination of the Extended Signal Tail. Because automated boundary estimation in heavy-tailed empirical distributions remains an approximation, the exact algorithmic mechanics and hyperparameter thresholds of this operational utility are fully detailed in Appendix A.

Furthermore, because double-centering shifts the coordinate origin to the centroid, it mechanically shrinks the trace of the NMI matrix. While this geometric transformation advantageously mitigates positive estimation bias, its direct algebraic consequence is that the average eigenvalue organically drops below 1.0. In high-dimensional environments, Random Matrix Theory dictates that unstructured finite-sample noise does not evaluate to zero; instead, it forms a continuous spectral distribution of idiosyncratic informational variance centered exactly around this matrix mean. Consequently, applying the classical Kaiser criterion (retaining $\lambda > 1.0$, which assumes the unit-diagonal of a correlation matrix) or an analogous mean-eigenvalue threshold $(\lambda > \text{Trace}/m)$ is flawed in regimes where N is not much larger than m. Rather than separating generative signal from idiosyncratic informational variance, this threshold arbitrarily fractures the continuous distribution, falsely extracting any idiosyncratic informational variance that naturally fluctuates above the average.

While this Dual-Diagnostic Ensemble provides a principled automated approach to estimating these phase transitions, it relies on empirical heuristics lacking exact Random Matrix Theory bounds. However, from a pragmatic engineering perspective, applying approximate heuristics to a structurally valid, non-linear metric space represents an objective upgrade over applying any standard threshold to a fundamentally distorted linear space.

Because real-world systems frequently exhibit complex internal hierarchies among correlated drivers – which can produce large internal informational variance drops independent of the idiosyncratic baseline – and because idiosyncratic informational variance distributions themselves contain unpredictable localized structures, automated extraction should ideally not operate as a blind black box. The Entropic Scree methodology dictates that when human oversight is possible, these algorithmic estimates serve as analytical baselines, which should be coupled with visual inspection of the log-linear spectral decay to formally verify both the primary generative boundary and the full extent of the Extended Signal Tail.

4 Five Tiers of Dimensionality

By evaluating the NMI spectral decay alongside the theoretical baseline, the Entropic Scree formally distinguishes between the true underlying drivers of a system and the linear approximations of standard statistics. It decouples the abstract notion of "Representational Rank" into five distinct tiers:

- **Intrinsic Generative Rank (r):** The intrinsic topological dimensionality of the system's true underlying drivers. However, the nominal mechanistic headcount (r) must be distinguished from the resolvable informational volume.

  If the r roots are themselves entangled, their unique, idiosyncratic signal is smaller than if they are uncorrelated. In sparse, finite-sample regimes, this unique signal is submerged by estimation noise, potentially causing the eigensolver to organically collapse entangled roots and map the system's *effective* generative rank. As sample size (N) increases and the idiosyncratic informational variance floor shrinks, the framework gains the statistical estimation power to identify r.

  This resolution is equally dependent on the physical breadth of the dataset (m). As established in recent data-architectural theory (Lee-St. John et al., 2026), expanding m acts as a high-resolution filter that physically resolves underlying degeneracy, separating overlapping probability distributions until distinct generative roots become observationally distinguishable. Thus, as with all rank estimation methods, successful extraction fundamentally relies on both sufficient N to overcome statistical noise and sufficient m to overcome structural ambiguity.

  Notably, this dynamic – where sufficient resolution (N and m) is required to split entangled roots – does not equally apply to the Entropic Scree's ability to map purely deterministic non-linear transformations. For strict monotonic bijections (e.g., odd-powered polynomials like X_1^5), the transformations share 100% mutual information with their root and are entirely robust to resolution dependency. While non-monotonic transformations (e.g., even-powered polynomials like X_1^2) induce a specific, structural loss of mutual information due to symmetric coordinate folding (a geometric mechanic detailed further in Appendix D), their shared probabilistic footprint remains dominant. Conversely, standard linear estimators fragment all such non-linearities into spurious orthogonal dimensions – an algorithmic failure that persists regardless of available sample resolution. Because the Entropic Scree evaluates shared probability mass rather than linear variance, it inherently recognizes the foundational footprint of both odd and even transformations, compressing these artifactual linear expansions back towards their single generative root.

- **Realized Intrinsic Root Expansion Rank (K_{rlzd}):** The discrete physical count of active mathematical terms (interactions and polynomials) expanding the intrinsic roots, acting as the "linear casting space" from which observable data is projected. Variance-accounting machines like standard PCA falsely perceive every dimension of this expanded casting space as an independent generative dimension.

- **Effective Intrinsic Root Expansion Rank (K_{eff}):** The continuous, variance-weighted effective volume of the K_{rlzd} expansion axes.

- **Observed Generative Rank (K_{roots}):** The discrete dimensional complexity of the empirical dependency structure. Provided the environment possesses sufficient redundant dependency to elevate the primary signal above the idiosyncratic informational variance, K_{roots} acts as an information-accounting compressor – compressing the fragmented linear expansions of K_{rlzd} back into a dense, non-linear map, serving as a robust empirical estimator for r.

- **Total Unique Probabilistic Volume (R_{eff}):** The dataset's total continuous probability volume. This aggregate encapsulates not only the unique signal volume capable of forming shared bivariate dimensions, but also the system's idiosyncratic informational variance, which consists of both random stochastic noise (Structural Uncertainty and independent measurement error) and unshared signal geometry (pure, higher-order root combinatorial fragments too isolated to form a shared bivariate dimension). Because it explicitly absorbs both these isolated signal fragments and the random static, R_{eff} natively inflates strictly above the Observed Generative Rank (K_{roots}).

Here, we distinguish between these underlying theoretical tiers and their empirical manifestation on the Entropic Scree. Because the proposed methodology relies on a bivariate (2D) mutual information matrix, it is structurally blind to pure higher-order multivariate synergies. When observing complex interactions, this bivariate filter shears a portion of the informational variance away from the primary generative roots. Consequently, the empirical eigenspectrum fractures into two distinct operational phases:

1. **Observed Generative Rank (K_{roots}) Cliff:** The obvious drop on the left side of the scree mapping directly to the underlying drivers and estimating r. This constitutes the actionable structural boundary for downstream feature extraction.

2. **Extended Signal Tail ($K_{extended}$):** An artifactual, extended shelf of eigenvalues following the primary roots. It is not an intrinsic tier of dimensionality, but a methodological byproduct containing the residual root expansion signal (the informational variance of interactions and folded polynomials geometrically sheared off the roots by the bivariate filter and binning artifacts).

While this Extended Signal Tail does not represent independent generative drivers and should never be extracted as such, its probabilistic mass is fundamentally deterministic. Therefore, it must be actively recaptured during signal-to-noise accounting. Doing so ensures that this structured, recoverable residual informational variance is properly credited to the generative roots, preventing it from being incorrectly relegated to the idiosyncratic informational variance floor alongside Structural Uncertainty, independent measurement error, and unshared signal geometry.

5 Quantifying Structural Gravity (AIG and FSIG)

Isolating the physical footprint of the extracted topological axes requires partitioning the dataset's Total Unique Probabilistic Volume (R_{eff}) against its absolute physical capacity. While the raw NMI matrix possesses a diagonal of exactly 1.0 (yielding an uncentered trace of m), the double-centering transformation shrinks the trace to $Tr(\mathcal{M}_c)$ and consumes one degree of freedom to establish the geometric origin, capping the active spectral rank at $m-1$. However, the physical dataset remains composed of exactly m observable variables. By evaluating the Total Unique Probabilistic Volume against this absolute physical baseline, we can execute a tripartite apportionment:

1. **Redundant Signal Volume ($m-R_{eff}$):** Idiosyncratic informational variance cannot share mutual information. By evaluating the exponentiated entropy, R_{eff} acts as a strict count of effective independent dimensions (variable equivalents). Although the underlying linear algebra of double-centering mathematically restricts the active spectral space to a maximum rank of $m-1$ and shrinks the matrix trace, the applied engineering environment requires accounting for exactly m physical columns. To bridge this spectral geometry back to physical reality, the framework makes a deliberate methodological choice: it partitions the volume against the absolute physical baseline (m) rather than the centered trace ($Tr(\mathcal{M}_c)$). The single degree of freedom consumed by double-centering represents the dataset's centroid. Because this centroid is a shared macroscopic property of the interconnected system, the framework methodologically classifies its consumed mass as redundant structural signal rather than discarded variance. Consequently, subtracting the Total Unique Probabilistic Volume (R_{eff}) from the absolute physical column count (m) explicitly isolates the redundant signal volume – capturing both the specific topological repetitions and the centroid.

2. **Unique Signal Volume:** Calculated by projecting the structural weight of the extracted signal onto the Total Unique Probabilistic Volume. To capture the valid expansion signal and prevent it from artificially inflating the idiosyncratic informational variance, this calculation must be integrated out to the Extended Signal Tail boundary ($K_{extended}$): $\left(\frac{\sum_{i=1}^{K_{extended}} \lambda_i^+}{m_+}\right) \times R_{eff}$.

3. **Idiosyncratic Informational Variance:** The remainder of R_{eff}. This residual bucket contains two theoretically distinct components that the eigensolver cannot geometrically separate: stochastic noise (Structural Uncertainty and independent measurement error), and unshared signal geometry (highly localized root combinatorial interactions that lack the gravitational mass to form a shared macroscopic dimension). Because the linear spectral extraction cannot distinguish between random stochastic error and entirely isolated geometric fragments, both fall past $K_{extended}$ into this idiosyncratic informational variance.

5.1 Average Informational Gravity (AIG)

By summing the unique and redundant signal volumes, the methodology captures the total shared signal volume. Dividing the comprehensive total by the Observed Generative Rank (K_{roots}) yields the Average Informational Gravity (AIG). AIG is a direct measure of *variable equivalents.*

Because AIG evaluates the total shared signal volume (which sweeps up the entire Extended Signal Tail) and divides it purely by the true number of roots, it acts as a geometric rebundling mechanism – it collects the residual expansion informational variance sheared off by the bivariate blindspot and distributes it back to the generative drivers. It serves as the framework's most robust macroscopic metric, describing how much of the dataset the roots control on average without being distorted by differential variance leakage. Because Jaccard normalization is conservative, AIG possesses a lower bound of exactly 1.0 for any valid geometric manifold.

5.2 Factor-Specific Informational Gravity (FSIG)

While Average Informational Gravity provides a useful global baseline, assuming this structural mass is distributed uniformly across all extracted factors contradicts the skewed topology of most complex systems. To map the physical strength of individual topological axes, Factor-Specific Informational Gravity ($FSIG_i$) projects a primary root's proportional spectral weight against the total shared signal volume:

$$FSIG_i = \left(\frac{\lambda_i^+}{\sum_{j=1}^{K_{roots}} \lambda_j^+} \right) \times \text{total shared signal volume}$$

While FSIG acts as an essential diagnostic for evaluating relative root dominance, practitioners must interpret its absolute magnitude with caution due to the informational variance leakage. If an underlying root is highly interactive within the dataset, a disproportionate amount of its informational variance will leak into the Extended Signal Tail due to the bivariate filter, artificially deflating its primary eigenvalue (λ_i^+). Conversely, an isolated root will retain nearly all of its variance on its primary axis. Because FSIG assumes eigenvalue mass perfectly maps to systemic importance, the differential leakage caused by the bivariate blindspot ensures that FSIG acts strictly as a conservative lower bound of a highly interactive root's true physical influence.

If the empirical phase transitions converge ($K_{roots} = K_{extended}$), it indicates one of two conditions: either the dataset lacks the systemic resolution (sufficient sample size N to lower the idiosyncratic informational variance, or sufficient predictor breadth m to aggregate the residual signal) to elevate the Extended Signal Tail out of the idiosyncratic informational variance bulk, or the system truly possesses a straightforward architecture devoid of highly entangled, non-monotonic combinations. Provided the environment possesses sufficient resolution to rule out signal suppression, this convergence diagnoses the absence of the bivariate blindspot. Because zero variance leaks into an Extended Signal Tail under these pristine conditions, the derived structural gravity metrics (AIG and FSIG) cease to act as conservative lower bounds and can be trusted as exact physical constants of the system's topological mass.

5.3 Structural Topology Profile

Normalizing the complete extracted signal against the primary axis yields the Structural Topology Profile ($\text{FSIG}_{1-K}/\text{FSIG}_1$). Because FSIG scales proportionally with spectral variance, this profile is algebraically equivalent to the raw eigenvalue ratio ($\lambda^+_{1-K}/\lambda^+_1$), translating the invariant geometry of the matrix into comparative physical footprints.

In systems where physical phenomena are determined by dense combinations of the underlying generative roots (e.g., a complex clinical event driven by the intersection of multiple physiological systems), the observable reality inherently constitutes an entangled web of shared informational variance. To map this structure, the eigensolver is forced to draw its primary axis through the densest core of shared information. This results in a disproportionately large FSIG_1 and a steep, immediate drop-off in the resulting profile, acting as a topological signature of a highly centralized network. From an engineering perspective, this warns practitioners that the dataset cannot be safely partitioned into isolated predictive models without severing the synergistic links inherently bound together in the underlying system.

Conversely, a perfectly flat profile (where FSIG_1 is strictly uniform with the subsequent valid factors) indicates a completely modular, disjointed topology, confirming the observable variables are genuinely isolated expressions of pure roots.

However, in heavily entangled environments where the underlying roots are symmetrically distributed (e.g., a balanced combinatorial mesh where no single root dominates the others), the profile will exhibit a distinct hybrid signature. The profile will display a sharp primary drop-off – mapping the centralized global hub of combinatorial overlap – immediately followed by a flat plateau across the remaining factors. This secondary plateau suggests that once the global entanglement is isolated, the underlying generative drivers exert nearly equal structural gravity across the environment. Furthermore, step-functions in this profile (e.g., where FSIG_1 and FSIG_2 are equally large, followed by a sharp drop) allow practitioners to diagnose multi-core centralizations – identifying datasets composed of two or more distinct, internally entangled macro-domains. By combining the Observed Generative Rank (K_{roots}) with this continuous mass distribution, this framework evaluates not merely the absolute count of underlying drivers, but the physical architecture of their real-world combinations.

6 Distinction Between Diagnostic Mapping and Feature Extraction

The application of eigenvectors ($\mathbf{V}$) extracted from the Normalized Mutual Information matrix differs fundamentally from standard PCA. In standard PCA, the extracted eigenvectors represent linear coefficients in the original coordinate space. Consequently, standard practice dictates projecting the raw dataset directly onto these axes via a simple linear combination ($\mathbf{X} \cdot \mathbf{V}$) to generate a reduced set of continuous latent features.

This direct projection is structurally invalid within the Entropic Scree framework. The framework operates entirely within topological information space, where the matrix evaluates joint probability mass rather than continuous physical magnitude. The resulting eigenvectors map the geometry of shared entropy – identifying exactly which clusters of variables co-move – but they do not constitute linear weights. Attempting to compute a dot product between a vector of raw, mixed-type physical observations and an informational eigenvector constitutes a category error, yielding a scalar devoid of interpretable meaning.

Consequently, the Entropic Scree should be utilized strictly as a diagnostic architectural blueprint. It operates as a structural baseline, estimating the Intrinsic Generative Rank (r) via the Observed Generative Rank (K_{roots}) and mapping the informational gravity (AIG/FSIG) of the system. However, the physical construction of the lower-dimensional manifold must be delegated to downstream, non-parametric manifold learning algorithms or non-linear autoencoders. By supplying these downstream extraction tools with the Observed Generative Rank (K_{roots}) identified by the Entropic Scree, practitioners can encourage empirical estimators to encode the true topological drivers, helping protect against the curse of dimensionality and the memorization of trailing idiosyncratic informational variance.

6.1 Interpreting Informational Loadings: Bipolar Modules and the Configurational Space

While direct linear projection ($X \cdot V$) is mathematically invalid, the extracted eigenvectors ($\mathbf{V}$) themselves remain interpretable as a map of the system's structural topology. Because the eigendecomposition operates on a centered inner-product space, the underlying structural relationship between any two variables is geometrically approximated by the product of their factor loadings.

This geometry determines how practitioners should interpret the extracted axes to identify discrete structural neighborhoods:

- **Variables with the same sign (A Localized Cluster):** If two variables possess high loadings with the exact same sign (e.g., both highly positive or both highly negative) on a given factor, their geometric inner product is strictly positive. As established in Section 3.2, a positive inner product indicates an acute angle, representing a shared dependency strictly greater than the pair's expected baseline. These variables form a highly connected, internally dependent structural module.

- **Variables with opposite signs (Structural Estrangement):** If two variables possess high loadings with opposite signs on the same factor, their inner product is large and negative. This indicates an obtuse angle extending across the dataset's centroid, mapping a shared dependency significantly below the systemic baseline.

In classical factor analysis, practitioners are trained to expect a strict one-to-one relationship between a factor and a variable cluster. Under linear assumptions, variables loading on opposite ends of a factor belong to the exact same mechanistic construct − they are simply inversely correlated (e.g., when one feature increases, the other deterministically decreases).

In the Entropic Scree framework, this traditional one-to-one intuition must be entirely inverted. Because the NMI matrix evaluates shared information rather than directional vectors, a negative inner product does not indicate an inverse mechanism; it indicates an absence of shared information. Variables loading with opposite signs represent structurally independent mechanisms that are estranged from one another.

Consequently, an eigensolver operating in this space will often extract *bipolar factors*. To capture maximum topological variance, the algorithm draws its axes across the widest voids in the data manifold, frequently anchoring a single geometric axis on two completely disconnected structural modules at its opposite poles.

This provides an efficient heuristic for baseline network extraction: by grouping variables with extreme positive loadings into one module, and variables with extreme negative loadings into a separate module, practitioners can cleanly untangle up to $2K_{roots}$ distinct, foundational sub-networks directly from the primary structural axes.

However, the physical identity of these extreme factor poles is dictated entirely by the observable distribution of the dataset. Because the eigensolver is constrained to maximize captured topological variance, it drops its primary anchors directly into the densest hubs of shared information:

- **In Modular Systems (Decentralized):** If the dataset predominantly consists of pure, isolated expressions of the underlying drivers, these isolated roots possess the highest structural gravity. Consequently, the extreme poles will anchor directly onto the isolated generative roots, while sparse combinatorial intersections are suspended diagonally in the multidimensional space between axes.
- **In Entangled Systems (Centralized):** In complex environments where the vast majority of observable variables are combinatorial intersections of multiple roots, these combinations act as dominant topological hubs. They share mutual information with multiple independent root networks simultaneously, exceeding the structural gravity of the isolated variables. In these regimes, the dominant combinations physically anchor the extreme poles of the primary axes. The "pure" root variables, lacking sufficient shared mass to anchor their own primary dimensions, are relegated to the topological periphery, loading moderately across multiple trailing factors.

Practitioners do not need to guess which topological regime they are operating within; it is explicitly diagnosed by the Structural Topology Profile ($\text{FSIG}_{1-K}/\text{FSIG}_1$) introduced in Section 5.3. A perfectly flat profile confirms a strictly modular system where the primary anchors can be safely interpreted as isolated roots. Conversely, a steep initial drop – indicating a disproportionately large primary axis (FSIG_1) – demonstrates that the system is highly centralized. Furthermore, if this massive primary drop is immediately followed by a flat plateau across the subsequent factors, it explicitly diagnoses a hybrid architecture: a heavily entangled global hub built upon a balanced, symmetrically distributed root system. In these entangled regimes, attempting to interpret the primary anchor variable clusters as representing pure, isolated generative roots constitutes a geometric category error. The anchors represent the boundaries of the dominant combinations, and downstream non-linear manifold unmixing is required to isolate the pure generative drivers.

6.2 Informational Boundaries and Rank Exhaustion

Although a complex dataset may contain hundreds of dominant combinations, the diagnostic restricts its primary structural extraction to exactly r dimensions (yielding $2r$ geometric poles), as demonstrated by the Observed Generative Rank (K_{roots}). This restriction is governed by the absolute limits of informational degrees of freedom defined by the Data Processing Inequality (DPI) (Cover and Thomas, 2006). By the proofs of DPI, while r latent roots can interact to form thousands of non-linear observable combinations, those downstream combinations cannot generate novel, independent entropy. Consequently, the entire entangled web of combinatorial hubs remains confined within an r-dimensional informational volume.

Because the Normalized Mutual Information matrix evaluates shared probability mass rather than linear variance, it accurately reflects this volumetric boundary. The eigensolver does not *a priori* know r; it simply casts a full coordinate space to decompose the matrix. However, because the structural combinations are strictly bounded by r degrees of freedom, the shared generative probability mass is physically exhausted after exactly r dominant axes. This exhaustion triggers the primary macroscopic phase transition (K_{roots}), cleanly separating the r valid structural dimensions from the Extended Signal Tail and unstructured idiosyncratic informational variance. To capture the maximum possible shared variance before this signal drops off, the algorithm stretches its valid axes to the absolute geographical limits of the dataset. Therefore, the $2r$ anchor clusters extracted by the methodology do not represent an exhaustive census of all combinations; rather, they represent the specific combinatorial hubs defining the extreme outer boundaries of the dataset's true r-dimensional informational manifold.

For a formal mathematical proof establishing the absolute theoretical upper bound of the Extended Signal Tail – including the exact geometric mechanisms that prevent both multi-way interactions and pure polynomial expansions from perfectly compressing into the primary roots – see Appendix D.

6.3 Topological Information Space: An Operational Analogy

An operational analogy illustrates the distinction between the original underlying generative space and the vectors the eigensolver actually manipulates.

Consider a dataset generated by exactly four independent biological root systems ($r = 4$): Cardiovascular, Respiratory, Neurological, and Muscular. We populate this dataset with 1,000 observable clinical events. While a few events are isolated expressions of a single root, the vast majority are combinations forming two distinct clinical macro-states: Cardiopulmonary events (Cardiovascular + Respiratory) and Neuromuscular events (Neurological + Muscular).

When this system is translated into the double-centered topological information space, the variables are not plotted by their physical magnitudes. Instead, they become vectors emanating from the dataset's centroid, where a variable's vector length (its double-centered diagonal value) dictates its total structural gravity, and the vector angle (derived from the double-centered inner product) dictates shared dependency. Cardiopulmonary events share extensive mutual information with one another, embedding them as a high-density vector bundle. Neuromuscular events form a second, distinct high-density bundle. An isolated Cardiovascular event shares less total information, forming a shorter, weaker vector.

The eigensolver, entirely blind to the original four underlying mechanisms, projects its primary axis through the widest geometric void in the manifold to maximize captured topological variance. It drops its extreme positive anchor directly into the Cardiopulmonary hub, and its extreme negative anchor into the Neuromuscular hub.

Critically, because mutual information strictly evaluates dependency rather than linear direction, this bipolar axis does not indicate an inverse relationship (e.g., as Cardiopulmonary events increase, Neuromuscular events deterministically decrease). Instead, it represents pure structural estrangement. The two combinatorial hubs share zero overlapping roots, meaning their pairwise mutual information evaluates to zero. In double-centered space, this absence of shared information generates a negative inner product, physically pushing the clusters to opposite poles of the exact same structural dimension. The primary axes are anchored exclusively by the geometric boundaries of these disjoint combinations, while the true, isolated generative roots lack the structural mass to anchor their own primary dimensions and are suspended weakly between them.

Furthermore, this analogy geometrically separates the Observed Generative Rank (K_{roots}) from the Total Unique Probabilistic Volume (R_{eff}). If the data were perfectly clean and fully mapped by a true r-way multivariate metric space, all variables would sit flush on an r-dimensional hyperplane. However, the bivariate blindspot shears combinatorial variance into the Extended Signal Tail, while Structural Uncertainty and independent measurement error project individual variables off this plane into thousands of independent, random spatial dimensions. This uncorrelated static surrounds the valid structural geometry with N-dimensional idiosyncratic informational variance. Provided the shared probability mass of the signal exceeds the idiosyncratic informational variance, the eigensolver will prioritize mapping the primary plane (K_{roots}) and its accompanying Extended Signal Tail. Only after the macroscopic signal is exhausted does the algorithm expend its remaining axes mapping the idiosyncratic informational variance. This demonstrates why R_{eff} – which evaluates the entire continuous mass of the roots, the tail, and the idiosyncratic informational variance – inflates strictly above the Extended Signal Tail boundary ($K_{extended}$).

6.4 Combinatorial Alignment and Topological Denoising

Because the entropic factors map combinatorial boundaries rather than isolated roots, this framework provides a geometric baseline for verifying variable assignments. In classical linear frameworks, if a combinatorial variable is driven by both Root A and Root B, the resulting cross-loadings are frequently flagged as noise. Practitioners are often forced to discard the variable or arbitrarily misassign it exclusively to Root A, effectively destroying its true physical reality.

In the Entropic Scree, if a variable is incorrectly assumed to belong to an isolated root cluster, but the underlying geometry reveals its topological distance is significantly closer to a combinatorial boundary, the metric space explicitly flags this structural misalignment. By re-aligning variables exclusively to the combinatorial anchors that govern their shared information, practitioners achieve a denoised map of the system's observable combinatorial macro-states (governed by K_{roots}). Only after this observable network is structurally denoised should downstream, non-linear manifold unmixing algorithms be deployed to safely reverse-engineer those combinations back into their isolated generative roots.

7 Testing Linear Sufficiency: The Dimensional Inflation Index

Standard Principal Component Analysis operates under an assumption of linearity, yet it possesses no internal diagnostic mechanism to verify if that assumption holds true for a given dataset. By formalizing the delta between classical variance-based rank and the observed informational rank, the Entropic Scree can be utilized not merely as a replacement for PCA, but as a formal diagnostic bounding box for PCA itself.

This evaluation is motivated by the theoretical relationship between rank under strict parametric conditions. If a dataset is continuous and generated by a purely linear multivariate Gaussian process, Mutual Information reduces to a deterministic function of the Pearson correlation coefficient. While the non-negativity of the NMI matrix alters the continuous probability mass of the eigenvalues, because this transformation is strictly monotonic in the underlying correlation, the discrete topological rank is expected to closely track between the two representations. This relationship holds exactly for large-sample, well-separated signal, but is only locally well-approximated near the idiosyncratic informational variance – precisely the boundary region where the diagnostic is most often applied. Under pristine linear conditions, both the standard correlation matrix and the NMI matrix should therefore produce a closely comparable count of eigenvalues that break out of the spectral tail.

Consequently, the Realized Intrinsic Root Expansion Rank (K_{rlzd}) estimated by the classical PCA elbow (K_{PCA}) and the Observed Generative Rank (K_{roots}) mapped by the Entropic Scree are expected to closely align under pristine linear conditions.

It is crucial to recognize that while the boundaries of both models can reflect methodological artifacts in complex systems, the origins of these artifacts are fundamentally distinct. The spurious dimensions comprising the PCA elbow (K_{PCA}) are absolute consequences of the model's core theoretical axioms: because standard PCA mandates a linear, orthogonal coordinate system, it must physically fabricate new dimensions to map non-linear combinations. Its dimensional inflation is a fundamental failure of its generative worldview. Conversely, the Extended Signal Tail observed in the Entropic Scree is not a failure of information theory, but merely a computational byproduct of applying a bivariate filter to multivariate synergies. Therefore, to accurately isolate the exact degree of spurious linear fragmentation, the diagnostic must compare PCA's theoretical failure directly against the true intrinsic manifold, entirely bypassing the Entropic Scree's computational tail.

This permits the formalization of the Dimensional Inflation Index (Δ_K):

$$\Delta_K = K_{PCA} - K_{roots}$$

Typically, K_{PCA} is derived directly from the visual or mathematical elbow of the classical PCA scree plot. In non-linear environments, this elbow frequently still manifests; however, rather than capturing the true intrinsic generative rank, it expands to map the spurious orthogonal dimensions generated by linear fragmentation. Furthermore, in heavily entangled, extreme high-dimensional regimes ($m > N$)—such as the empirical simulation detailed below—standard PCA can fail to produce any structural elbow whatsoever, instead yielding a smooth, continuous spectral decline. Under these severe conditions, K_{PCA} should be estimated using the default linear heuristic (e.g., the Kaiser criterion, $\lambda > 1.0$) to quantify the unconstrained dimensional extraction. Evaluating Δ_K provides a practical, quantifiable heuristic for assessing linear sufficiency prior to executing downstream feature extraction:

- **Convergence ($\Delta_K \approx 0$):** The absence of dimensional inflation is consistent with an underlying dependency structure well-approximated by a simple linear factor model, suggesting spurious orthogonalization is negligible and that classical PCA is likely sufficient for the dataset.
- **Divergence ($\Delta_K \gg 0$):** A positive divergence quantifies the exact discrete severity of dimensional inflation (Section 2). A large delta – such as the $\Delta_K = 5,742$ observed in our empirical simulation below (Section 8) – highly suggests the presence of heavily non-linear synergies, mixed-data shape mismatches, or discrete topological states.

This suggests standard PCA is likely inappropriate as an extraction framework and motivates the use of non-linear manifold learning architectures.

8 Empirical Demonstration

To empirically validate the theoretical resolution of these linear constraints (Section 2), a highly non-linear, high-noise synthetic environment was constructed specifically designed to violate the assumptions of standard variance-based estimators. Furthermore, to demonstrate that the Entropic Scree provides structural diagnostics unavailable to existing non-linear machine learning baselines, it is evaluated alongside Standard PCA, Spearman Rank PCA, and Kernel PCA (RBF).

A system was generated driven by $r = 20$ continuous generative roots. To provide an unambiguous ground truth, these roots were deliberately constructed to be strictly orthogonal. This ensures their effective informational rank perfectly aligns with their discrete mechanistic count ($r = 20$), allowing us to formally isolate the dimensional inflation caused solely by non-linear expansions, free from the confounding effects of root-level correlations.

8.1 Generative Mechanics and Combinatorial Topology

These 20 roots were expanded into a pool of non-linear transformations (cross-interactions up to the 5th order and polynomials up to the 4th order). This 5th-order complexity forged a global design matrix containing a Realized Intrinsic Root Expansion Rank of exactly $K_{rlzd} = 21,759$ unique mathematical terms.

To simulate a heavily entangled, decentralized network, $m = 20,000$ observable proxy variables were generated. To establish the baseline generative structure, 5% of these proxies were reserved as pure idiosyncratic informational variance (i.e., static noise), leaving 19,000 actively entangled proxies. Every entangled proxy strictly sampled exactly 10 of the 20 base roots (50% root sparsity).

This uniform root sparsity dictates the macroscopic topology of the dataset. With 20 roots taken 10 at a time ($\binom{20}{10}$), there are exactly $184,756$ possible foundational blueprints. Because the simulation generates only 19,000 entangled proxies, it actively samples a mere $\sim 10.3\%$ of the available configurational space. This produces two topological properties: first, nearly every proxy possesses a unique generative blueprint; second, the dataset achieves a high degree of decentralization, with the 19,000 proxies scattered across a much larger combinatorial space. Nevertheless, this 10-root sampling also defines a universal overlap floor. The probability of any two random proxies sharing exactly k underlying roots follows a Hypergeometric distribution:

$$P(X = k) = \frac{\binom{10}{k}\binom{10}{10-k}}{\binom{20}{10}}$$

Evaluating this distribution reveals that the expected baseline overlap between any two proxies in the dataset is exactly 5 shared roots. Over 98% of all variable pairs share between 3 and 7 roots, embedding a nearly universal web of positive mutual information.

While the proxies are highly decentralized across the 184,756 root configurations, the 20 underlying roots remain overwhelmingly anchored. Because every proxy selects exactly

10 out of 20 roots, every driver has exactly a 50% chance of being included in a proxy's blueprint. By the Law of Large Numbers, each of the 20 generative roots is supported by an aggregate of approximately $10,000$ unique proxy vectors, providing structural density for the downstream algorithms to leverage.

Once a proxy's 10-root blueprint is established, it accesses a localized menu of mathematically aligned main effect and expansion terms. From this restricted menu, the proxy samples 10% of the available terms, assigning term weights drawn from a standard normal distribution ($\mu = 0$). Table 1 details how this geometry heavily biases toward high-order combinatorics, with Main Effects comprising a fractional minority of the design space.

This uniform 10% sampling heavily limits the observable frequency of highly complex interactions. A 1-root Main Effect only requires its single specific root to be selected by the proxy (50% probability). Thus, a specific Main Effect is drawn approximately 950 times across the matrix ($19,000 \times 0.50 \times 0.10$). Conversely, a specific 5-Way interaction requires all 5 of its specific roots to be simultaneously drawn (a mere $\sim 1.6\%$ probability). Consequently, any specific 5-Way interaction appears only ~ 31 times ($19,000 \times 0.016 \times 0.10$). This combinatoric bottleneck dictates exactly how much signal mass is available to crystallize in the extended manifold.

Table 1: **Design Matrix and Term Representation.** Comparison of term types within the global universe vs. an average local 10-root menu (approx. 667 terms).

Term Type	**Total in Global Universe ($K_{rlzd} = 21,759$)**	**Actual % in Local Proxy Menu**
Main Effects (1D)	20 (0.09%)	1.50% (10 terms)
Pure Polynomials (1D)	60 (0.28%)	4.50% (30 terms)
2-Way Interactions	190 (0.87%)	6.75% (45 terms)
3-Way Interactions	1,140 (5.24%)	17.99% (120 terms)
4-Way Interactions	4,845 (22.27%)	31.48% (210 terms)
5-Way Interactions	15,504 (71.25%)	37.78% (252 terms)

Finally, to simulate a sample-starved, high-noise environment, the dataset was restricted to $N = 10,000$ historical observations ($m > N$). Structural Uncertainty was injected directly into the generative manifold (SNR = 10.0), and independent measurement error was heavily applied to the resulting proxies (a continuous Signal-to-Noise Ratio of 2.0, and a 15% random bit-flip rate for binary variables).

8.2 Failure of Standard and Non-Linear Baselines

Standard PCA: First, because the global non-linear expansion ($K_{rlzd} = 21,759$) exceeds the sample size ($N = 10,000$), standard PCA hits a hard algebraic rank ceiling. It is mathematically forced to truncate the manifold at exactly $N - 1 = 9,999$ dimensions, permanently blinding it to the full geometric bounds of the root expansion space. Second, standard linear methods are incapable of mapping higher-order combinations back to their generative sources. Because non-linear interactions and polynomials are geometrically orthogonal to their constituent main effects in a linear coordinate system, standard PCA perceives the thousands of active terms as independent variables. This fractures the 20 true generative roots into thousands of spurious orthogonal dimensions. This dimensional inflation physically manifests in the PCA spectral decay (Figure 1, Panel 1): the spectrum exhibits absolutely no macroscopic phase transition, acting as a smooth, featureless tail that falls steadily out to the $N-1$ ceiling. With no visual elbow to rely on, the standard Kaiser criterion falsely extracts 5,762 principal components and entirely submerges the true structural rank.

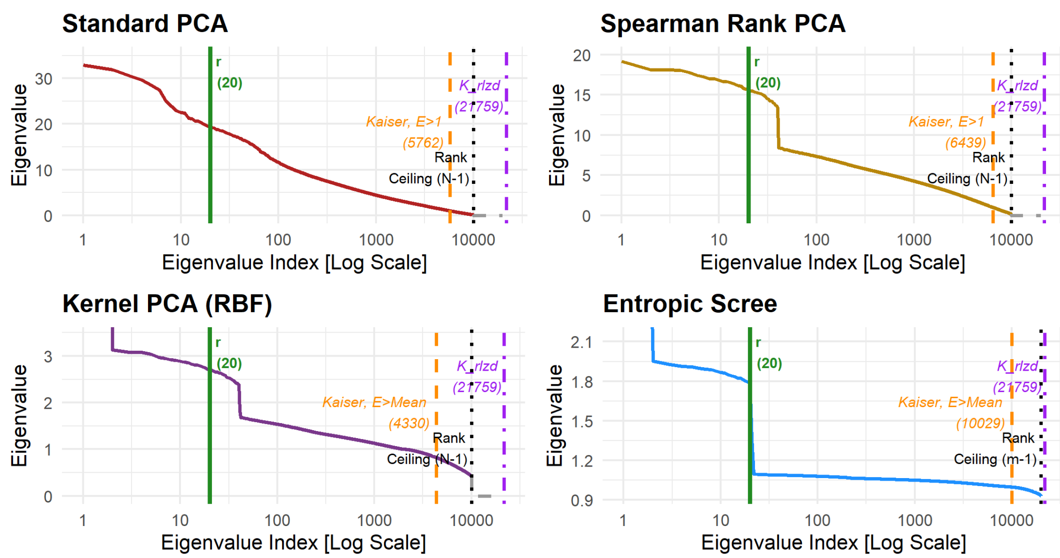

Figure 1: **Baselines vs. Entropic Scree.** *Panel 1 (Standard PCA):* The linear matrix fractures all expansions, yielding a smooth decay to the sample-size ceiling. *Panel 2 & 3 (Spearman & Kernel PCA):* Rank-based and RKHS projections exhibit a structural drop at index 40, reflecting the aggregation of the 20 true roots and their 20 folded even-polynomial counterparts into dominant dimensions. Because the multivariate cross-interactions lack monotonic correlation (Spearman) or spatial alignment (KPCA) with their parent roots, they are treated as independent variables. However, due to their sampling sparsity, they lack the shared variance to form distinct principal components, instead distributing their variance into an elevated, continuous tail. *Panel 4 (Entropic Scree):* Mutual information evaluates dependencies at their root, compressing both polynomials and all cross-interactions to reveal the Intrinsic Generative Rank ($r = 20$).

Spearman Rank PCA: To ensure the observed inflation is not merely an artifact of simple monotonic scaling, PCA was applied to the dense-ranked dataset. While rank transformations perfectly correlate strict monotonic bijections (e.g., pure odd polynomials) with their generative roots, they do not preserve correlation for non-monotonic geometry. Even polynomials (e.g., X^2 and X^4) symmetrically fold the probability space, yielding near-zero rank correlation with their base root. However, because X^2 and X^4 are monotonically correlated with each other, the eigensolver aggregates all even polynomials into a single orthogonal dimension per root.

This algorithmic aggregation concentrates statistical variance. Because each proxy independently samples terms, aggregating the linear root (X) with its odd polynomials (X^3) compounds their sampling frequencies, effectively doubling their representation within the covariance matrix. The even polynomials aggregate similarly. This combined variance generates exactly 40 dominant 1-dimensional axes: the 20 true roots and their 20 folded even-polynomial counterparts. Consequently, the Spearman eigenspectrum (Figure 1, Panel 2) exhibits a distinct structural drop exactly at index 40.

Beyond these 1D terms, the rank-based solver evaluates the multivariate cross-interactions. Because a cross-interaction (e.g., X_1X_2) is not monotonically correlated with its parent roots, the rank-based matrix evaluates it as an independent, orthogonal variable. However, due to the hypergeometric sampling constraints (Section 8.1), these sparse terms lack the shared variance required to form discrete principal components. Instead, their variance pools into a highly elevated, continuous tail. While there is a visible spectral drop following index 40, the eigenvalue at index 41 remains highly inflated (~ 8.0). From this elevated position, the spectrum merely forms a smooth, featureless tail that falls steadily out to the $N-1$ ceiling. Because this post-drop tail is so elevated and fails to flatten into a typical spectral floor, it is arguable whether the break at index 40 constitutes a valid structural elbow at all. This leaves a practitioner attempting a visual scree test with ambiguity: extracting 40 dimensions represents a best-case, 100% dimensional inflation, while arbitrarily slicing through the elevated tail with the Kaiser criterion ($\lambda > 1.0$) falsely extracts 6,439 components.

Kernel PCA (RBF): Kernel PCA represents the standard machine learning baseline for non-linear manifold mapping. By employing the kernel trick, KPCA bypasses the K_{rlzd} expansion wall and projects the data into a Reproducing Kernel Hilbert Space (RKHS) (Figure 1, Panel 3). While the RBF kernel accommodates continuous 1D non-linearities, it aligns the spatial trajectories of odd polynomials with the true roots and clusters the symmetrically folded even polynomials into independent dimensions. This aggregates their structural variance into 40 dominant axes, resulting in a visible structural drop following index 40. However, just as with Rank PCA, the eigenvalue at index 41 remains substantially elevated (exceeding 1.6), albeit to a lesser degree than in the Rank PCA result. Owing to the same hypergeometric sampling sparsity, the multivariate cross-interactions (2-Way through 5-Way) lack the spatial density to form isolated axes. Instead, their variance distributes across an elevated, continuous tail that falls steadily from index 41 onward. Because this continuous decline starts from such an elevated position rather than flattening into a typical spectral floor, interpreting the drop at 40 as a definitive elbow is questionable. This leaves standard diagnostic extraction compromised: a generous visual reading overestimates the rank at 40, while applying a mean-eigenvalue threshold arbitrarily slices through the tail to falsely extract 4,330 components.

8.3 Entropic Scree Extraction and Hypergeometric Crystallization

Applying the Entropic Scree methodology to this same dataset circumvents these limitations.

As outlined in Section 3, the double-centering transformation enforces a strict zero-sum constraint designed to mitigate positive finite-sample estimation bias. This structural

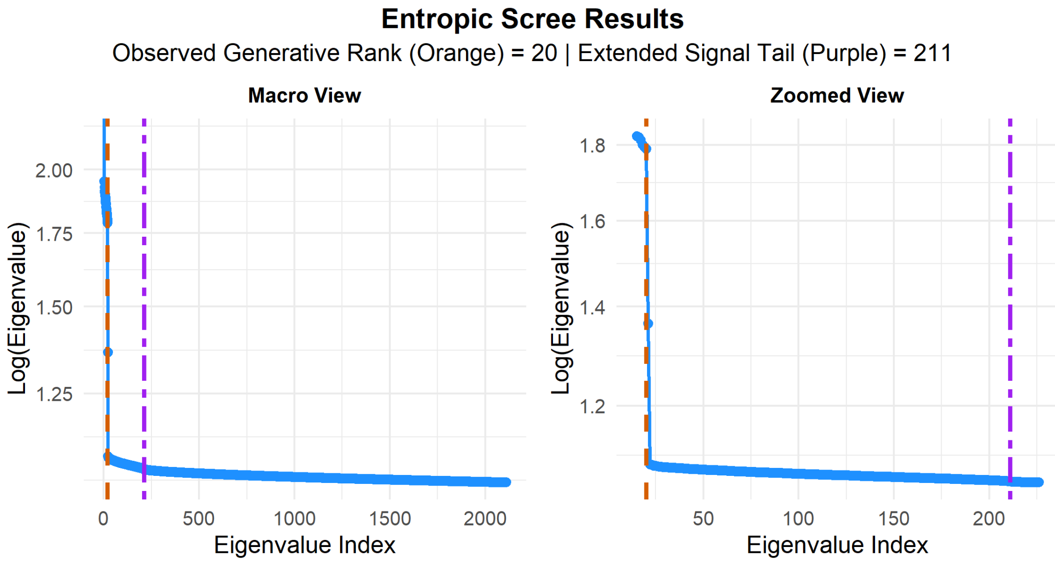

Figure 2: **Entropic Scree Extraction.** The macro view illustrates the global decay of the dataset's Total Unique Probabilistic Volume (R_{eff}). The zoomed view isolates the macroscopic phase transitions, revealing the Observed Generative Rank breaking out at $K_{roots} = 20$, followed by the dampened Extended Signal Tail.

correction is directly observable in the dataset's geometric baseline, explicitly shrinking the initial physical column count of $m = 20,000$ down to a Centered Trace ($Tr(\mathcal{M}_c)$) of 19,945.14. Because Jaccard normalization bounds the topological volume, non-Euclidean artifacts were entirely negligible (SCDR = 0.000%).

When parsing the Entropic Scree's decay (Figure 2), the automated Dual-Diagnostic Ensemble identified a rigid structural phase transition. Bypassing the primary axis ($FSIG_1$), the Log-Gap engine correctly identified the Observed Generative Rank at $K_{roots} = 20$. In stark contrast to the heavily elevated, ambiguous tails of the Rank and Kernel PCA baselines, the Entropic Scree's informational variance drops cleanly and definitively – with the primary residual eigenvalue at index 21 falling sharply to ~ 1.1. Scanning backwards from the deep idiosyncratic informational variance baseline with a 20-point linear window, the Triple-Tap engine estimated the termination of the Extended Signal Tail at $K_{extended} = 211$, triggered by a large 39.11σ breakout from the expected idiosyncratic informational variance trajectory.

This estimated Extended Signal Tail closely mirrors the physical footprint of the hypergeometric combinatoric bottleneck applied to the global design matrix. Because the Normalized Mutual Information matrix acts as a bivariate filter, the primary overlapping probability mass of the combinations is successfully compressed into the Intrinsic Generative Rank ($r = 20$). Furthermore, because pure polynomials are strictly univariate non-linearities, their variance is overwhelmingly absorbed by these primary roots, relegating their microscopic residuals directly into the idiosyncratic informational variance floor. However, the bivariate filter inherently shears off the unique synergistic variance of the cross-interactions, leaving behind orthogonal residuals. The global universe contains exactly 190 foundational 2-Way interactions. Because these specific low-order configurations possess the highest hypergeometric baseline representation across the dataset, their sheared bivariate residuals aggregate enough structural mass to crystallize into distinct geometric axes. Appending these 190 2-Way residuals to the 20 primary roots yields a theoretical capacity of 210 structural axes, tightly aligning with the Extended Signal Tail's empirical termination at index 211. Conversely, once complexity reaches the 1,140 3-Way interactions and beyond, the hypergeometric bottleneck becomes too severe. This extreme sampling sparsity strips their residuals of shared probabilistic mass, causing the residuals of the 3-Way, 4-Way, and 5-Way interactions to melt seamlessly into the unstructured idiosyncratic informational variance floor. It is important to emphasize that this specific termination at index 211 is not a universal mathematical bound of the NMI matrix, but a direct reflection of the 10% sampling sparsity engineered into this specific simulation. This alignment demonstrates the framework's capacity to accurately map the physical limits of an environment's observable combinatorial sparsity, rather than imposing an artificial theoretical ceiling.

The Total Unique Probabilistic Volume of the dataset ($R_{eff} = 19,963.87$) captures the system's entire continuous footprint. By evaluating this effective volume against the system's absolute physical capacity ($m = 20,000$), the methodology estimates that an overwhelming 98.549% of the environment consists of idiosyncratic informational variance. The active underlying shared signal accounts for a mere 1.451% of the topological volume.

Applying the structural gravity metrics outlined in Section 5, the extracted signal yields an Average Informational Gravity (AIG) of 14.512 variable equivalents. By integrating the entire probabilistic mass of the Extended Signal Tail into the total shared signal volume and rebundling it purely across the true roots, this metric formally rescues the combinatorial variance sheared off by the bivariate filter. Disaggregating this rebundled mass via Factor-Specific Informational Gravity reveals the hierarchical topology of the extracted manifold. Because the 5-root baseline overlap is universally shared, the first factor acts as a centralized gravitational hub, exerting a dominant footprint of $FSIG_1 = 74.546$ variable equivalents.

However, normalizing the extracted mass against this primary axis yields the system's Structural Topology Profile: $[1.000, 0.160, 0.158, 0.157, 0.156 \ldots 0.146]$. This dual-state profile perfectly diagnoses the hybrid architecture of the simulation. The distinct drop from the primary anchor directly maps the highly centralized network of overlapping combinations—specifically, the universal 5-root baseline overlap structurally embedded into the dataset. Yet, after isolating this global entanglement hub into Root 1, the subsequent 19 roots form a nearly flat plateau. Having successfully sheared away the macroscopic centralization, the eigensolver maps the symmetric, independent informational variance of the remaining factors across the subsequent dimensions, each controlling a uniform footprint of approximately 11.5 variable equivalents. This visually confirms that despite the 5th-order non-linear distortion and an environment blanketed by over 98.5% unstructured idiosyncratic informational variance, the heavily entangled combinations are built upon a decentralized, democratically distributed network of perfectly orthogonal roots.

Ultimately, these results demonstrate that the Entropic Scree successfully retrieved the Intrinsic Generative Rank ($r = 20$) where both classical linear estimators and non-linear baselines degraded. Evaluating the Dimensional Inflation Index reveals an extreme divergence ($\Delta_K = 5,742$), explicitly quantifying the total failure of linear correlation. Meanwhile, Rank and Kernel PCA yielded, at best, a 100% dimensional inflation – a liberal estimate confounded by their elevated and ambiguous tails (particularly evident in the Rank PCA spectrum). Together, these baseline failures provide evidence that established extraction paradigms are structurally invalid for environments driven by deep combinatoric synergies.

Table 2: **Empirical Simulation Summary.** Comparison of dimensional diagnostics extracted from a noisy, highly entangled ($m > N$) mixed-data environment.

Metric	**Observed Value**
Panel A: Environment Configuration	
Sample Size (N)	10,000
Intrinsic Generative Rank (r)	20
Observable Proxies (m)	20,000
	(80% Continuous, 20% Binary)
Independent Measurement Error Injected	SNR = 2.0 (Cont.), 15% Bit-flip (Bin.)
Structural Uncertainty Injected	SNR = 10.0 (Round 1 Noise)
Algebraic Rank Ceiling	9,999
Realized Intrinsic Root Expansion Rank (K_{rlzd})	21,759
Effective Intrinsic Root Expansion Rank (K_{eff})	3,519.33
Panel B: Standard PCA Diagnostics	
Kaiser Criterion: $\lambda > 1.0$	5,762
Visual Elbow (K_{PCA})	None (Smooth Decline; falls back to Kaiser)
Panel C: Spearman Rank PCA Diagnostics	
Kaiser Criterion: $\lambda > 1.0$	6,439
Visual Elbow	~ 40 (Ambiguous; obscured by elevated tail)
Panel D: Kernel PCA (RBF) Diagnostics	
Kaiser Equivalent: $\lambda > \text{Trace}/m$	4,330
Visual Elbow	~ 40 (Ambiguous; obscured by elevated tail)
Panel E: Entropic Scree Diagnostics	
Centered Trace ($Tr(\mathcal{M}_c)$)	19,945.14
Synergistic Curvature Deficit Ratio (SCDR)	0.000%
Kaiser Equivalent: $\lambda > \text{Trace}/m$	10,029
Observed Generative Rank (K_{roots})	20
Extended Signal Tail Rank ($K_{extended}$)	211
Total Unique Probabilistic Volume (R_{eff})	19,963.87
Idiosyncratic Informational Variance	98.549%
(Structural Uncertainty and $\perp$ Measurement Error)	
Total Shared Signal Volume (Unique + Redundant)	1.451%
Automated Extraction Method	Dual-Diagnostic Ensemble
Topological Variance Drop (Log-Gap Magnitude)	1.31× (23.8% Drop)
Triple-Tap Sigma Breakout	39.11σ
Average Informational Gravity (AIG)	14.512 (Tail Volume Rebundled)
Dominant Axis Gravity ($FSIG_1$)	74.546 Variable Equivalents
Secondary Axes Gravity ($FSIG_{2-20}$)	$\sim 11.91 - 10.92$ Variable Equivalents
Structural Topology Profile ($\text{FSIG}_{1-20}/\text{FSIG}_1$)	$[1.00, 0.16, 0.15 \ldots 0.14]$
	(Hybrid: Hub + Flat Plateau)
Dimensional Inflation Index (Δ_K)	5,742 (Severe Linear Divergence)

8.4 Robustness to Root Entanglement

While the strict root orthogonality of the primary simulation provides an essential baseline to isolate non-linear dimensional inflation, real-world generative drivers are frequently characterized by collinearity and entanglement. To prove the framework's robustness to this phenomenon, the simulation was additionally subjected to a sensitivity analysis across varying degrees of generative root entanglement.

As detailed above, Standard PCA systematically failed even with root orthogonality due to linear fragmentation. While the rank-based and RKHS baselines yielded a best-case, liberal estimate of a 100% dimensional inflation under these ideal orthogonal conditions, their elevated tails, particularly that observed in the Rank PCA result, precluded definitive extraction. However, as demonstrated in Appendix C, introducing even mild root entanglement completely collapsed these ambiguous structural elbows as well, rendering their eigenspectra structurally indistinguishable from the failed standard PCA spectra. Conversely, the Entropic Scree maintained a rigid, near-invariant extraction highly consistent with the orthogonal baseline.

9 Conclusion

The Entropic Scree replaces the variance-based paradigms of classical PCA with information-theoretic geometry. By evaluating Normalized Mutual Information within a Jaccard metric space, it successfully resolves the artificial deflation of mixed-data shape mismatch and completely bypasses the algebraic rank ceiling that traps standard linear models in sample-starved ($m > N$) environments.

By recognizing the deterministic redundancy of non-linear interactions, it successfully compresses the fragmented linear dimensions of the expansion space back down to the Intrinsic Generative Rank of the system. Furthermore, empirical demonstrations confirm that while standard linear, rank, and RKHS estimators systematically degrade under even mild root collinearity, the information-theoretic geometry of the Entropic Scree remains near-invariant to root entanglement.

Beyond resolving these linear artifacts, this framework defines a system's true complexity not by an unobservable headcount of mechanistic parts, but by its effective informational footprint. The Entropic Scree functions as a non-parametric, distribution-free diagnostic for any mixed-type dataset, and its comparative advantage over established extraction paradigms scales with system complexity – thriving in high-dimensional environments where the sheer volume of combinatorial expansions severely fragments baseline methods.

While all empirical estimators are universally bound by the resolution limits of finite sample sizes and predictor breadth, this framework ensures that complex non-linearities are compressed towards their root rather than fragmented into spurious dimensions. Furthermore, it formally acknowledges that "statistical power" in this context is a dual function: it requires sufficient sample depth (N) to minimize independent measurement error, and sufficient predictor breadth (m) to resolve root degeneracy (Lee-St. John et al., 2026). Ultimately, what the methodology extracts is not a physical parts list, but an operational map of what the environment possesses the statistical and structural power to resolve. If a multitude of entangled root causes generates only a few distinct axes of mutual information, those active macro-states constitute the true operational reality of the dataset.

Because this manuscript introduces the foundational theoretical architecture of the Entropic Scree, exhaustive empirical validation falls outside its current scope. Furthermore, because the current framework evaluates a strictly bivariate (pairwise) metric space, it natively maps overlapping synergies but remains structurally blind to pure higher-order multivariate dependencies (e.g., XOR configurations) where marginal pairwise information evaluates to zero. While the synthetic stress-test successfully demonstrates its theoretical mechanics in ill-conditioned environments ($m > N$), establishing its absolute boundaries as a generalized diagnostic requires rigorous future benchmarking.

Subsequent research should execute systematic ablation studies mapping the algorithm's sensitivity across extreme sample-to-feature ratios (N/m), varying signal-to-noise distributions, differing proportions of continuous versus discrete variables, and alternative discretization heuristics (e.g., Sturges (Sturges, 1926) or Freedman-Diaconis (Freedman and Diaconis, 1981)). Specifically, while empirical ablation demonstrates that the extraction of the Observed Generative Rank (K_{roots}) is invariant to discretization bias, the continuous gravity metrics (AIG, FSIG, and the Structural Topology Profile) remain sensitive to the finite-sample error introduced by binning resolution. Validating the real-world stability of these gravity metrics – and formalizing optimal discretization strategies to standardize their absolute volumetric estimates – necessitates application to standard, high-dimensional empirical environments, such as genomic arrays, single-cell RNA-sequencing, and complex, mixed-type clinical datasets (e.g., MIMIC-III). By formally defining the Jaccard-normalized metric space and the log-linear phase transition mechanics, this paper provides the theoretical scaffolding required to support those future applied investigations.

Code Availability

The original R implementation – including the core Entropic Scree function (v1.0.0) and the exact scripts used to generate the simulated data and results presented in Section 8 and Appendix B – is publicly available at: `https://github.com/tjleestjohn/Entropic-Scree`.

Native R and Python implementations via standard package managers will be released shortly to facilitate immediate deployment and broad accessibility.

- **R:** `install.packages("Entropic.Scree")` *(Coming Soon)*
- **Python:** `pip install Entropic-Scree` *(Coming Soon)*

Computational Efficiency and Scaling

To ensure scalability across high-dimensional environments, the provided implementation resolves the computational bottlenecks traditionally associated with large-scale mutual information estimation. This efficiency is achieved through three architectural optimizations:

1. **Pre-Processing Optimization:** The pipeline leverages highly optimized, vectorized data structures across both the Python and R environments to rapidly purge invariant columns, perfect collinearity, and non-linear monotonic duplicates prior to matrix computation.
2. **Bypassing Density Estimation (Robust Discretization):** Rather than relying on computationally prohibitive continuous density estimators (such as k-nearest

neighbors), the algorithm maps continuous variables to discrete integer states via equal-frequency (quantile) binning and dense-ranking. To strike an optimal balance between preserving non-linear structural resolution and mitigating finite-sample entropy bias, the methodology utilizes a dynamic heuristic that scales the number of discrete bins proportionally to the cube root of the sample size ($N^{1/3}$) (Scott, 1979).

While discretization inherently results in the loss of intra-bin variance, the diagnostic stability of the Entropic Scree is structurally dampened against extreme sensitivity to this specific hyperparameter choice. Because the methodology employs Jaccard normalization (Section 3), the resulting NMI metric evaluates a ratio: if alternate binning rules (e.g., $N^{1/2}$ or Freedman-Diaconis (Freedman and Diaconis, 1981)) increase the marginal entropies, both the mutual information and the exact joint entropy denominator scale proportionally. Consequently, the normalized geometric distances between variables are expected to remain highly stable, preserving the relative topology of the eigenspectrum. Furthermore, because K_{roots} is identified via the relative, logarithmic phase transition of the eigenspectrum rather than absolute variance magnitudes, the structural oracle is robust to the uniform trace scaling that plagues unnormalized information matrices, safely transforming complex multivariate probability estimation into highly efficient integer frequency counting.

However, while this conservative discretization protects the extraction of the discrete structural rank (K_{roots}), it deterministically impacts the Total Unique Probabilistic Volume (R_{eff}). By acting as a low-pass filter that deliberately erases weak micro-dependencies to suppress finite-sample error, the $N^{1/3}$ heuristic guarantees that the derived structural gravity metrics (AIG and FSIG) function as mathematically conservative **lower bounds** of a factor's true continuous footprint. This algebraic dynamic is fully detailed in Appendix B.

3. **C++ Integration and Multi-Threading:** The core computational bottleneck – the pairwise calculation of joint entropies for the $m \times m$ matrix – bypasses the high-level interpreters entirely. Both the Python and R distributions delegate this operation to custom C++ backend engines that dynamically parallelize frequency calculations across multiple CPU cores using `OpenMP`, executing natively in RAM.

Declaration of Competing Interests

The author is affiliated with Enli. The methodology and codebase introduced in this manuscript are provided entirely open-source for the scientific community. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors, and the author declares no other competing financial or personal interests.

Appendix A. Automated Extraction Heuristic Mechanics

The automated baseline utility operationalized in the accompanying codebase employs a Dual-Diagnostic Ensemble to map the system's structural phase transitions (K_{roots} and $K_{extended}$). It is designed specifically to suppress heavy-tailed edge fluctuations in sparse regimes and identify the macroscopic drop in topological variance. Let $\lambda_1^+ \geq \cdots \geq \lambda_m^+$ denote the positive clipped eigenvalues of the double-centered NMI matrix $\mathcal{M}_c$.

Phase 1: Continuous Bulk Boundary Detection (The Macroscopic Cliff)
As established in Section 4, while all non-linear root interactions formally belong to the Extended Signal Tail, sparse interactions that lack the shared probability mass to overcome Structural Uncertainty and independent measurement error melt into the idiosyncratic informational variance floor. Because this unstructured variance forms a dense, continuous spectral distribution, evaluating algorithmic gaps deep within this space risks triggering on random stochastic density fluctuations rather than valid structural phase transitions. To prevent this, the algorithm establishes a macroscopic search boundary, separating the discrete dimensional space from the continuous bulk.

1. Define the search space of trailing eigenvalues whose magnitude falls below the mean centered trace.

2. Calculate the absolute differences (gaps) between successive eigenvalues in a localized window deep within this trailing idiosyncratic informational variance bulk (defaulting to the bottom 5% of active eigenvalues). Let $\Delta_{\text{bulk\_base}}$ be the maximum absolute gap within this deep continuous tail.

3. The top of the idiosyncratic informational variance bulk ($\text{Index}_{\text{boundary}}$) is identified as the first instance (scanning backward from the tail) where an absolute gap exceeds a conservative structural multiplier of this baseline: $\Delta_i > \tau \times \Delta_{\text{bulk\_base}}$, where $\tau = 20$.

If no gap satisfies this threshold, the boundary defaults to the Kaiser criterion equivalent ($\text{Index}_{\text{boundary}} = |\{\lambda_i^+ > \frac{Tr(\mathcal{M}_c)}{m}\}|$).

Phase 2: Dual-Boundary Extraction
Operating entirely within the bounded signal space ($i \leq \text{Index}_{\text{boundary}}$), the algorithm deploys two independent mathematical engines, each designed to map a distinct operational boundary of the non-linear manifold:

Engine A: Maximum Secondary Eigenvalue Ratio (Log-Gap)
This engine isolates the Observed Generative Rank (K_{roots}), identifying the primary gravitational drops associated with the true latent drivers.

1. The primary gap ($i = 1$) is explicitly excluded to bypass the primary topological axis ($FSIG_1$), forcing a minimum manifold extraction of $K \geq 2$.

2. The structural elbow is formally estimated by maximizing the logarithmic gap between successive eigenvalues:

$$K_{roots} = \underset{2 \leq i \leq \text{Index}_{\text{boundary}}}{\text{argmax}} \left(\ln(\lambda_i^+) - \ln(\lambda_{i+1}^+) \right)$$

This integer represents the actionable extraction boundary to be passed to downstream non-parametric encoders.

Engine B: Triple-Tap (Dynamic Linear Breakout)
While the Log-Gap evaluates relative magnitude, the Triple-Tap engine evaluates geometric trajectory deviation to confirm the phase transition out of the continuous bulk. It maps the absolute edge of the Extended Signal Tail ($K_{extended}$), capturing the configurational variance sheared off the roots by the bivariate filter.

1. **Topological Stitching:** To prevent macro-gaps from fracturing the regression geometry, the void separating the shared signal from the idiosyncratic informational variance bulk is mathematically stitched. If a large gap is detected, the boundary coordinate is artificially elevated (setting the baseline to the top of the bulk plus a localized multiplier of $\Delta_{\text{bulk\_base}}$) to ensure continuous geometry.

2. **Rolling Regression:** Scanning backward from the top of the idiosyncratic informational variance bulk, a localized linear regression (default 20-point window) is continuously fitted to model the expected geometric trajectory and local standard error (σ) of the idiosyncratic informational variance tail.

3. **Dynamic FWER Breakout:** To rigorously separate shared signal from idiosyncratic informational variance without relying on arbitrary hardcoded thresholds, the engine dynamically scales the required breakout multiplier to enforce a targeted Family-Wise Error Rate (FWER). By applying a Bonferroni correction across the active search space (targeting a system-wide false positive rate of $\alpha = 0.01$), the algorithm computes a dynamic critical t-value based on the localized degrees of freedom. The engine identifies $K_{extended}$ at the exact index where the empirical eigenvalue diverges from the expected trajectory by a magnitude exceeding this dynamic t-scaled threshold, confirming a statistically significant structural crystallization out of the continuous bulk. While this dynamic FWER breakout provides a robust threshold for automated extraction, it must be acknowledged as an empirical heuristic designed to approximate the phase transition, rather than a formally proven Random Matrix Theory bound.

To ensure complete topological accounting, both engines operate in a dynamic discovery mode, natively scanning past the average baseline trace to map subtle interaction configurations. However, the methodology permits the strict enforcement of the continuous trace boundary ($\text{Index}_{\max} = |\{\lambda_i^+ > \frac{Tr(\mathcal{M}_c)}{m}\}|$) functioning as a dynamic Kaiser equivalent when conservative estimation is preferred. Ultimately, these heuristics provide analytical baselines that should be coupled with visual inspection of the log-linear spectral decay.

Appendix B. Robustness to Discretization Bias (Binning Ablation Study)

While the Entropic Scree successfully maps structural topology by transforming continuous distributions into discrete probability mass, this reliance on empirical probability estimation introduces a theoretical vulnerability: the finite-sample estimation error. As established by the Miller-Madow correction, empirical entropy and mutual information estimators exhibit a strictly positive bias that scales proportionally with the marginal cardinalities of the interacting variables. In practice, this suggests that discretizing continuous variables into a higher number of bins artificially inflates the pairwise mutual information, potentially causing highly-binned idiosyncratic informational variance to swell and obscure the true macroscopic phase transition.

To prove the Entropic Scree framework is structurally immune to this discretization bias, a controlled ablation study was executed on the synthetic environment. The continuous proxies (80% of the dataset) were subjected to three highly distinct discretization regimes designed to intentionally manipulate their marginal entropies:

1. $N^{1/3}$ **(Default - 22 bins):** A conservative, equal-frequency heuristic that mitigates data sparsity.
2. $N^{1/2}$ **(Dense - 100 bins):** A high-cardinality binning rule explicitly designed to trigger Miller-Madow bias and inflate the idiosyncratic informational variance bulk.
3. **Freedman-Diaconis (Extreme - 208 bins):** A data-driven heuristic scaling bin width dynamically via the Interquartile Range (IQR). In this specific heavily-tailed synthetic environment, it acted as a hyper-dense extreme.

The complete $10,000 \times 20,000$ system was re-evaluated under each regime.

Table 3: **Discretization Ablation Results.** The primary generative rank remains invariant despite manipulation of the continuous binning targets. However, the artificial compression of the Total Unique Probabilistic Volume (R_{eff}) inflates the total shared signal volume, which deterministically shifts the continuous structural gravity metrics (AIG and FSIG).

Discretization Rule	**Global Bin Target**	**Total Unique Probabilistic Volume** (R_{eff})	**Shared Signal Volume** (%)	**Observed Generative Rank** (K_{roots})	**Average Info. Gravity** (AIG)	**Primary Factor Gravity** ($FSIG_1$)
$N^{1/3}$	22	19,963.87	1.45%	20	14.51	74.55
$N^{1/2}$	100	18,996.07	6.21%	20	62.12	1,068.64
Freedman-Diaconis	208	16,727.77	38.07%	20	380.69	7,298.46

As demonstrated in Table 3, imposing a substantial increase in the cardinality of the continuous variables behaves exactly as information-theoretic theory predicts: the finite-sample estimation error inherently inflates pairwise mutual information across the matrix. This artificial noise makes the variables appear more highly correlated, mechanically driving down the dataset's Total Unique Probabilistic Volume (R_{eff}). However, despite this global compression of the independent probability space, the Observed Generative Rank remained invariant at $K_{roots} = 20$ across all conditions.

This structural invariance provides supportive evidence for the architecture of the Entropic Scree framework. Because the matrix normalizes the shared information strictly against the exact Joint Entropy of the interacting pair, the normalization mechanism acts as a dynamic algorithmic stabilizer. When extreme discretization (such as the 208-bin Freedman-Diaconis threshold) artificially inflates the raw mutual information via finite-sample noise, the marginal entropies constituting the joint denominator inflate proportionally. The resulting metric evaluates the ratio of shared non-linear volume rather than its absolute magnitude. Furthermore, because the structural elbow is evaluated via relative logarithmic phase transitions rather than absolute variance magnitude, the framework mechanically flattens the heteroskedastic noise tiers and preserves the relative topological geometry of the macroscopic phase transition.

Importantly, visual inspection of the eigenspectra across these ablation regimes (Figure 3) confirms that while finite-sample estimation error alters the idiosyncratic informational variance bulk, it cannot bypass the Jaccard normalization protecting the Observed Generative Rank. As additional discretization (100 and 208 bins) injects more finite-sample error into the continuous variables, the entire idiosyncratic informational variance floor elevates and homogenizes into a heavier continuous tail. However, despite this aggressive upward shift, the true Intrinsic Generative Rank ($r = 20$) remains sharply distinct and structurally preserved. This provides empirical evidence that the diagnostic successfully separates invariant generative signal from volatile, hyperparameter-dependent artifacts.

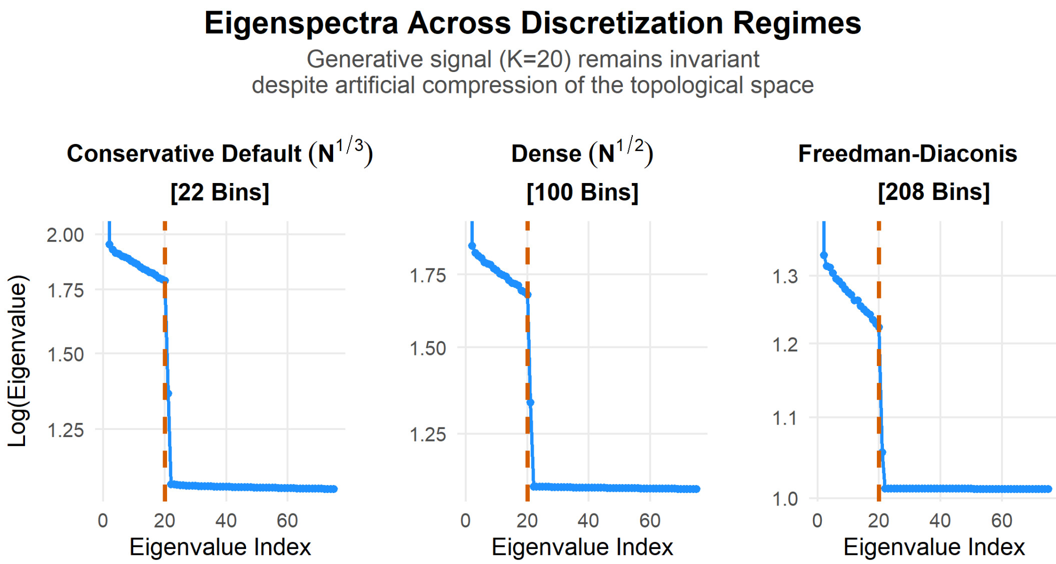

Figure 3: **Eigenspectra across Discretization Regimes.** As the continuous bin count increases from 22 (left) to 100 (center) to 208 (right), the injected finite-sample estimation error elevates and homogenizes the idiosyncratic informational variance into a smooth, heavy tail. Despite this artificial compression of the topological space, the true Intrinsic Generative Rank ($r = 20$) remains structurally distinct.

Consequently, while the discrete diagnostic extraction of the Observed Generative Rank (K_{roots}) is highly robust to discretization density, the continuous mass distribution is not. Because finite-sample estimation error injects artificial shared variance across the entire matrix, dense discretization causes the primary topological axis ($FSIG_1$) to act as a spectral sponge, disproportionately absorbing system-wide finite-sample error and artificially compressing the Total Unique Probabilistic Volume (R_{eff}). Conversely, the conservative $N^{1/3}$ heuristic deliberately acts as a low-pass filter, successfully suppressing this noise at the cost of erasing true, fine-grained micro-dependencies (via the Data Processing Inequality). By rendering weak dependencies invisible, the conservative heuristic artificially flattens the trailing eigenspectrum, inherently overestimating R_{eff}.

The net impact of this volume overestimation on structural gravity is deterministic. While an inflated R_{eff} artificially increases the unique signal volume ($P_{sig} \times R_{eff}$), it simultaneously decreases the redundant signal volume ($m - R_{eff}$). Because total shared signal volume

(TSV) combines these as $TSV = m - R_{eff}(1 - P_{sig})$ – where the signal proportion P_{sig} is strictly less than 1 – the loss in redundant volume always overpowers the gain in unique volume. Consequently, an overestimation of R_{eff} results in an underestimation of the total shared signal volume. Average and Factor-Specific Informational Gravity (AIG and FSIG) must therefore be interpreted as strictly conservative lower bounds rather than absolute physical constants when the conservative $N^{1/3}$ heuristic is utilized. They represent the minimum guaranteed structural footprint of a factor, deliberately stripped of the volatile micro-dependencies erased by the low-pass filter. Therefore, while the discrete rank remains invariant, practitioners cannot safely compare continuous gravity metrics across different environments unless a standardized, conservative binning heuristic is enforced to control this bidirectional estimation bias.

Appendix C. Robustness to Root Entanglement

The primary empirical simulation (Section 8) utilized orthogonal roots to establish an unambiguous mathematical ground truth for dimensional inflation. However, real-world generative drivers are rarely perfectly orthogonal; they are frequently characterized by collinearity and entanglement. To validate the framework's stability under these conditions, the identical simulation configuration ($m = 20,000$, $N = 10,000$, 5th-order complexity) was re-evaluated across varying degrees of latent root entanglement (Low, Medium, and High). To objectively control this collinearity, the generative correlation matrices were constructed using a standard random correlation matrix generator, modulating the Dirichlet distribution parameter (α_d) (Joe, 2006; Lewandowski et al., 2009) to 15.0, 0.75, and 0.001 to induce increasing systemic entanglement.

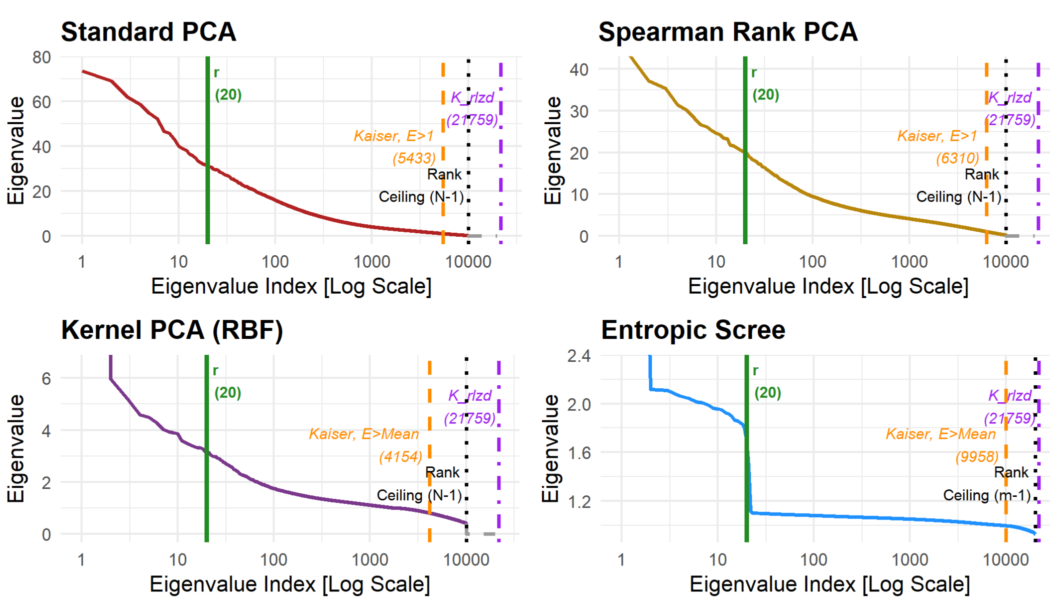

Figure 4: **Eigenspectra: Low Entanglement ($\alpha_d = 15.0$).**

As established in Section 8.2, Standard PCA failed to produce any structural elbow in the complex non-linear simulated environment, even when roots were orthogonal. It fractured non-linearities into a large number of spurious orthogonal dimensions. Furthermore, while the simple orthogonal baseline permitted Spearman Rank PCA and Kernel PCA (RBF) to cluster symmetrically folded polynomials into dominant axes, interpreting the resulting structural drop at index 40 as a definitive elbow represents a liberal estimate (yielding a 100% dimensional inflation at best). Because their post-drop eigenvalues remained significantly elevated above a typical spectral floor, traditional threshold extraction was already confounded, particularly in the Rank PCA context.

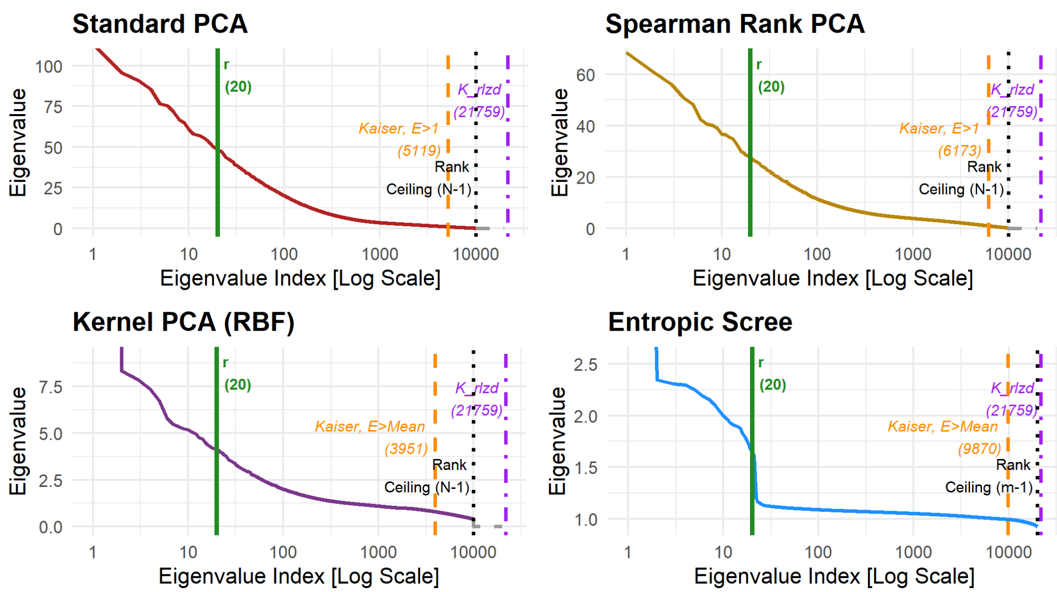

Figure 5: **Eigenspectra: Medium Entanglement ($\alpha_d = 0.75$).**

When the 20 roots were subjected to entanglement, these non-linear baselines systematically degraded across all tested severities. At the very first introduction of root collinearity (Low Entanglement, Figure 4), the Rank and Kernel PCA eigenspectra lost their elbows (from the orthogonal root context) at 40 entirely. Instead, they displayed a large eigenvalue drop between indices 2 and 3. Because this artifactual cliff overshadows all subsequent signals, standard automated heuristics – such as the Maximum Secondary Eigenvalue Ratio (Log-Gap) – are triggered prematurely at this early drop. Following this false elbow, the spectra exhibited a smooth, featureless tail that fell steadily all the way to their respective $N-1$ algebraic rank limits, permanently obscuring the remaining generative drivers. Similar failure patterns were observed in the spectra at Medium (Figure 5) and High (Figure 6) Entanglement levels.

This degradation is a direct consequence of their reliance on orthogonal variance. When the roots are even mildly correlated, linear solvers and RKHS projections disproportionately aggregate the vast majority of the shared variance into the first few principal components. Rather than dropping to a distinct structural floor, however, the remaining independent

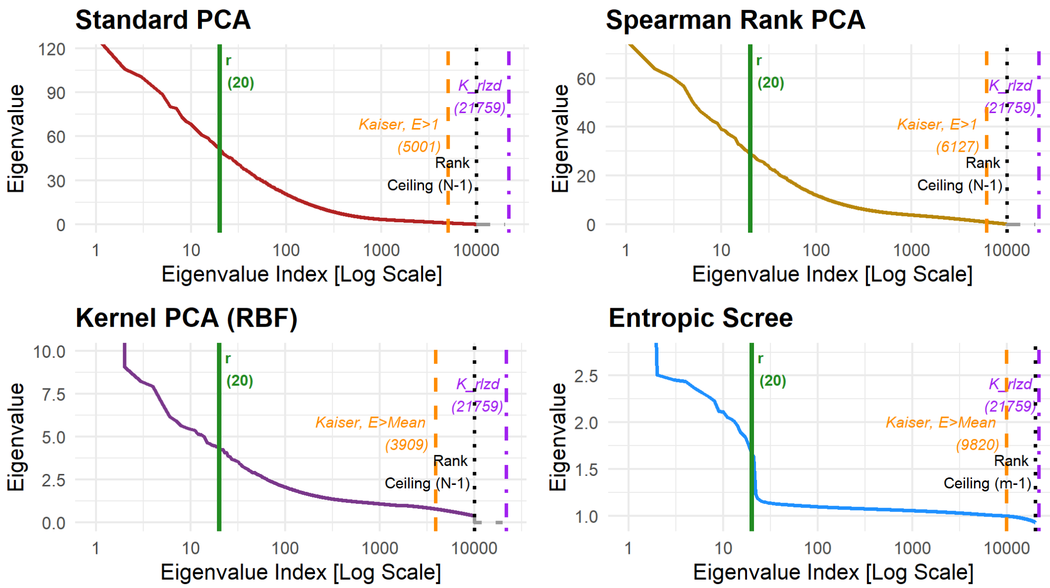

Figure 6: **Eigenspectra: High Entanglement ($\alpha_d = 0.001$).**

variance of the subsequent true roots is reduced to the same spectral tier as the inflated, high-order combinatorial noise (the non-primary $21,739$ expansion terms). Because the solvers cannot distinguish the remaining independent signal of the true roots from this highly elevated combinatorial static, the structural elbow is erased. Instead of a discrete phase transition, the spectrum degrades into a smooth, continuous tail that begins high above the typical spectral floor and falls steadily, rendering valid extraction impossible.

Conversely, the Entropic Scree proved highly robust to root entanglement across all conditions. Because the methodology evaluates shared probability mass (Normalized Mutual Information) rather than orthogonal spatial variance, it does not require the underlying roots to be independent to map the boundaries of the manifold. Even in the highest entanglement condition, the automated Dual-Diagnostic Ensemble identified the macroscopic phase transition at index 21, remaining exceptionally consistent with the true Intrinsic Generative Rank ($r = 20$) extracted in the orthogonal baseline. This confirms that while even mild root collinearity structurally blinds variance-based estimators by forcing true generative signals to smear into the combinatorial tail, the information-theoretic geometry of the Entropic Scree holds a rigid, invariant boundary.

As theorized in Section 4, extreme root entanglement possesses the potential to submerge unique idiosyncratic signals, potentially causing the observable rank to collapse into a lower *effective* generative rank. However, in this empirical simulation, the Entropic Scree successfully maintained its extraction at the true intrinsic rank ($r \approx 20$) across all entanglement conditions without collapsing. This stability directly demonstrates the resolution mechanics established in Section 4: because the environment possessed massive structural breadth ($m = 20,000$) and adequate sample depth ($N = 10,000$), the framework retained the statistical and geometric power required to physically resolve the underlying degeneracy. While the linear and RKHS baselines collapsed into a general factor vacuum, the information-theoretic geometry successfully separated the highly entangled probability distributions to preserve the true generative headcount. Nevertheless, as the correlation between two discrete generative drivers approaches unity (1.0), their mutual information converges, and their joint probability mass becomes functionally indistinguishable. Under conditions of near-perfect collinearity, the Entropic Scree will eventually – and correctly – collapse these drivers into a single, shared informational dimension, explicitly mapping their true, unified effective rank.

Appendix D. Theoretical Bounds of the Extended Signal Tail

As established in Section 4, the Entropic Scree inherently compresses the fragmented linear dimensions of the expansion space. However, because the bivariate filter and empirical discretization heuristics leave residual structural footprints, it is necessary to mathematically define the absolute theoretical upper bound of the Extended Signal Tail ($K_{extended}$).

From a pure accounting perspective, the absolute maximum theoretical limit for the Extended Signal Tail is exactly equal to the Realized Intrinsic Root Expansion Rank (K_{rlzd}). This represents the discrete headcount of every active non-linear combination and pure polynomial in the dataset. While the eigensolver compresses overlapping combinations, the mathematical requirement for reaching this absolute linear ceiling rests on the generation of unique, orthogonal residuals.

Bivariate Residuals of Multi-Way Interactions

Because the Normalized Mutual Information (NMI) matrix operates as a bivariate filter, it successfully captures the pairwise overlapping probability mass of complex multi-way combinations (e.g., $X_1X_2X_3$), compressing their shared variance into a small number of dense topological hubs. However, the unique synergistic effect of that specific multi-way interaction is sheared off. If the specific combinatorial nature of the interaction produces even a microscopic sliver of unique, unshared variance that is distinct from every other combination, that residual represents an orthogonal vector in the double-centered space. For the signal to stretch to the absolute limit of K_{rlzd}, it simply requires that after all shared overlap is accounted for, every single cross-interaction retains at least a fractional orthogonal residual that the eigensolver is forced to map.

Incomplete Compression of Pure Polynomials

It might intuitively follow that pure polynomials (e.g., X^2, X^3) should perfectly compress into the Observed Generative Rank (K_{roots}), allowing them to be legally subtracted from the K_{rlzd} ceiling. While rank-based and RKHS baselines artificially fuse these dimensions (as observed in Section 8.2), within a structurally rigorous information-theoretic space, perfect subtraction is mathematically invalid due to two distinct physical mechanisms:

1. **Folding Effects (Even Polynomials):** Even polynomials (e.g., X^2, X^4) applied to zero-centered distributions are non-monotonic; they fold the probability distribution, structurally destroying information (specifically, the sign bit). While X perfectly determines X^2, knowing X^2 cannot perfectly predict X. Consequently, X^2 possesses strictly less entropy than X ($H(X^2) < H(X)$). Because Joint Entropy evaluates to

the larger variable ($H(X, X^2) = H(X)$), the resulting Jaccard Normalization ($\frac{H(X^2)}{H(X)}$) evaluates to strictly less than 1.0. The polynomial lacks the resolution to map the lost sign bit, creating a geometric shape mismatch. The eigensolver is obligated to generate an orthogonal residual axis to map this specific folded variance.

2. **Binning Artifacts (Odd Polynomials):** Odd polynomials (e.g., X^3, X^5) are strictly monotonic bijections. In continuous theory, no information is folded, and their Jaccard NMI is exactly 1.0, which would yield perfect structural compression. However, generating the empirical matrix requires finite-sample discretization. Applying equal-frequency binning physically stretches the polynomial curve, changing the local density of the data points and causing rigid threshold collisions. This discretization error artificially fractures the perfect bijection, forcing the NMI slightly below 1.0 and leaving an artifactual sliver of unique variance.

Because even polynomials fundamentally leave structural residuals (due to folding), and odd polynomials leave artifactual residuals (due to binning), they cannot be subtracted from the linear headcount. Every single active term contributes at least a fractional orthogonal residual to the double-centered space. Therefore, while empirical compression typically exhausts the expansion mass drastically earlier than the absolute linear ceiling, the final trace of valid residual signal is mathematically capped at exactly K_{rlzd}.

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